Mandatory Disclosure and Female Representation in Corporate Leadership: Evidence from NASDAQ

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Abstract

We study how firms and financial markets respond to mandates requiring disclosure of information. The context is NASDAQ's 2021 board diversity rule, a comply-or-explain regulation requiring listed firms to disclose board diversity statistics and either appoint at least two diverse directors (one female and one underrepresented minority or LGBTQ+; one diverse director for boards of five or fewer) or explain why they do not have the required diversity. Unlike gender quotas, this disclosure-based mandate operates through market pressures rather than financial penalties, allowing reputational concerns to shape firm behavior. Using NYSE-listed firms as a control group, we find that mandatory disclosure increases female board representation by approximately 4%, with effects concentrated among larger firms and those with higher pre-regulation Environmental, Social, and Governance ratings. Event-study evidence shows abnormal returns of approximately 0.75% for NASDAQ firms around regulatory approval and approximately -1.3% around judicial vacatur, with larger effects for firms with all-male boards. The positive valuation effects appear consistent with the regulation reducing information asymmetries. Firms that choose to explain most often cite supply-side constraints such as an inability to find qualified candidates. Our findings indicate that mandatory disclosure generates modest increases in gender diversity in corporate leadership.

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1 Introduction

How do firms and financial markets respond to information disclosure mandates? A new regulatory frontier has emerged to increase diversity in corporate boardrooms: mandates that require disclosure of diversity statistics and explanations for non-diverse boards. Unlike quotas, mandatory disclosure affects behavior by altering information available to market participants, potentially generating market pressures that can influence firm conduct (Stiglitz 2002; Leuz and Wysocki 2016). If firms face sizable reputational costs from stakeholders for revealing a lack of diversity, they may add women to their boards to avoid negative scrutiny, rather than merely providing an explanation. Alternatively, if information surrounding board diversity was already widely available to stakeholders prior to the regulation, perhaps through voluntary disclosures (e.g., Ross 1979; Grossman 1981), then mandatory disclosure regulations should have limited impacts. On the financial side, if the board of directors solves a constrained optimization problem, then imposing additional reputational costs, to the extent they are applicable, should lower valuation measures. However, standardized reporting required by mandatory disclosure may mitigate information asymmetries, increase liquidity, and therefore improve financial outcomes.

In this paper, we provide evidence on the effects of mandatory diversity-disclosure regulations in the United States. We find that mandatory disclosure increases female board representation by approximately 4% and that financial markets reacted positively to the rule's approval. Our setting is the NASDAQ board diversity rule, approved by the Securities and Exchange Commission (SEC) on August 6, 2021. The rule required NASDAQ-listed firms to disclose the annual gender and demographic composition of their boards and either (i) include at least one female and one other diverse director (one director who self-identifies as an underrepresented minority or LGBTQ+), or (ii) explain why they did not.[1] The regulation featured a multi-year phase-in period (one diverse director by December 31, 2023 and two by December 31, 2025 or 2026 depending on listing tier)

[1]Boards with five or fewer members needed only one diverse director. Smaller reporting companies and foreign issuers could meet the two-diverse-director objective with two female directors. The rule imposed no financial penalties for companies choosing to explain. We focus on gender diversity in this paper because data on board gender composition is more complete and reliably measured than information on ethnic, racial, or sexual orientation diversity.

and was ultimately struck down by the Fifth Circuit Court in a 9-8 decision on December 11, 2024. We answer four sets of questions: 1) Do firms increase gender diversity in response to the mandatory diversity-disclosure requirement, and what types of firms are most responsive? 2) How do investors perceive the regulation as reflected by share price reactions to its adoption and subsequent repeal? 3) How does the regulation affect medium-run financial outcomes? and 4) How do firms that maintain non-diverse boards explain their choice, based on a textual analysis of their explanations?

To evaluate the causal effects of mandatory disclosure on annual measures of board diversity and financial outcomes, we compare NASDAQ-listed firms (treated) with NYSE-listed firms (control) using a difference-in-differences design. NYSE-listed firms serve as an ideal control group because they are subject to similar U.S. regulations enforced by the SEC but are not subject to the NASDAQ-specific mandate to disclose diversity. Although the two exchanges differ cross-sectionally, the identifying assumption relies on parallel trends and not identical baseline characteristics. As a robustness check, we use the synthetic difference-in-differences (SDiD) estimator of Arkhangelsky et al. (2021), which constructs a counterfactual for NASDAQ firms as weighted combinations of NYSE firms that best reproduce pre-treatment trends.[2] We use firm fixed effects and cluster standard errors at the firm level.

We find mandatory disclosure increases the female board share by 0.8 percentage points – approximately a 4% increase relative to a baseline mean of about 20% (Figure 1). The increases in the female share are concentrated among larger firms and those with higher Environmental, Social, and Governance (ESG) scores, suggesting these firms may have faced greater reputational costs for disclosing a lack of diversity. Compliance occurs primarily at the extensive margin, consistent with all-male boards adding a single female director to avoid filing an explanation. The SDiD estimator yields qualitatively similar results.

In standard asset-pricing frameworks, news about expected firm value is reflected in share

[2]As a robustness check, we also estimate a triple-difference specification using variation between NASDAQ and NYSE firms, before and after the regulation, and among companies that did and did not have diverse boards prior to the regulation.

prices at the time it arrives (MacKinlay 1997). We examine Cumulative Abnormal Returns (CARs) around both the SEC's approval of NASDAQ's diversity rule and its subsequent vacatur by the Fifth Circuit. Following the rule's approval, NASDAQ-listed firms experienced larger positive abnormal returns than NYSE-listed firms, with gains concentrated among firms that previously had no female directors. The two-day cumulative abnormal returns (CAR) after SEC approval is 0.748% for NASDAQ firms and statistically significant at conventional levels under several test statistics, compared with a negligible CAR for NYSE firms. In contrast, the rule's judicial vacatur is associated with a negative CAR for NASDAQ firms (-1.307%) and negligible abnormal returns for NYSE firms, with the losses again concentrated among firms that lacked board diversity prior to the court's decision.[3] We highlight the null effects for NYSE firms, which mitigate the usual concern about confounding events.

We provide suggestive evidence that the regulation reduced information asymmetries by requiring standardized reporting. First, the valuation effects are smaller for firms with high "Big 3" institutional ownership prior to the regulation where information surrounding board diversity may have already been available.[4] Second, the regulation appears to have lowered the effective spread, increasing liquidity in the market. In terms of other outcomes, in the three years following the regulation, we find small or insignificant effects on firm operating performance and key governance outcomes, including CEO turnover and compensation. We find no systematic evidence of evasive behavior, such as delistings or shifts in IPO venue. Firms did, however, adjust board composition at the margin: boards became more educationally credentialed and included more first-time directors, though we observe no reductions in other forms of relevant experience. Textual analysis of explanations provided by firms that remained non-diverse indicates that supply-side constraints such as a lack of qualified candidates and merit-based hiring considerations constitute the majority

[3] As a robustness check, we also compute abnormal returns at the portfolio level (Jaffe 1974; Allen and Wahid 2023). Under this specification, we do not detect significant abnormal returns around SEC approval, while the judicial vacatur is associated with negative abnormal returns of about -0.6% for all NASDAQ firms and about -1.3% for NASDAQ firms with all-male boards; no discernible effects are observed for NYSE firms.

[4] We find the first-stage point estimate on female share is smaller when controlling for "Big 3" institutional ownership, suggesting information on diversity may have been more widely available among these firms. This result contrasts with the finding in Hu, Hung, and Li (2025) in the Canadian context that first-stage effects are larger for firms with high institutional ownership.

of stated rationales. Taken together, our findings indicate that mandatory diversity-disclosure generated market responses consistent with reduced information asymmetries while inducing modest changes in board composition, highlighting both the promise and limits of disclosure-based governance.

Relationship to the Literature: Our paper contributes to three strands of literature. It is most directly related to recent research studying the effects of mandatory disclosure of diversity statistics. Bakke et al. (2021) and Hu, Hung, and Li (2025) study Canada's 2014 policy that required mandatory disclosure of female representation on corporate boards. Bakke et al. (2021) find that Canada's regulation increased the female share by 4 percentage points or approximately 40%. From examining share prices, they find that firms most affected by the regulation exhibit positive and statistically significant cumulative abnormal returns around the announcement, with two-day abnormal returns of 2%. Hu, Hung, and Li (2025) find similar first-stage point estimates and show that the increase in the percentage of women directors is larger when firms face stronger shareholder oversight, as reflected by ESG-friendly investors, pension fund ownership, and institutional ownership. In addition, they show that firms with stronger diversity commitments experience a greater increase in ownership by ESG-friendly investors.[5]

Unlike the Canadian context, we document substantially smaller effects of NASDAQ's mandatory disclosure regulation on female representation. Our evidence does not support the view that the "naming and shaming" effect of comply-or-explain regulations has similar effects on board composition as quotas, as argued by Fried (2021). Several factors may contribute to the smaller point estimates in our study. First, this regulation did not bundle requirements to disclose diversity statistics with requirements to disclose specific policies for increasing female representation.[6] Second, the U.S. corporate governance model is associated with shareholder rather than stakeholder

[5] See also Bourveau, Gao, and Hope (2025), who study Canada's follow up 2020 regulation that requires disclosure of racial diversity as opposed to gender diversity. They find racial diversity on corporate boards increases by roughly 33% relative to baseline levels, but document minimal effects for racial diversity in senior management.

[6] As discussed in Bakke et al. (2021), the Canadian regulation required firms to disclose the details of any policies concerning the identification and nomination of women directors, the board's consideration of the representation of women in the director identification and selection process, whether the firm has adopted targets for representation of women on the board, and whether the firm has director term limits. The amendment is principles-based, as listed firms are required to disclose these policies or to provide an explanation for their absence.

primacy, as articulated by Friedman (1970). Third, the time period also plays a factor – board gender diversity doubled from 2010 to 2020 in the U.S., and Diversity, Equity, and Inclusion (DEI) policies in the workplace faced backlash over our sample period (Bian, Li, and Li 2023). Despite these differences, we similarly find that ESG-oriented firms particularly increased female representation and that investors appear supportive of mandatory disclosure. Given the small changes to financial outcomes following the regulation, we share the interpretation of Hu, Hung, and Li (2025) that investors' positive reaction to mandatory disclosure may reflect changes to expected returns rather than cash flows. Relative to prior work, our study is also unique in providing a textual analysis of comply-or-explain justifications in the diversity setting.

Research on DEI in the workplace has largely focused on gender quotas on corporate boards. While their effects on firm performance remain debated, there is broad consensus that quotas substantially increase female board representation. For example, studies of Norway's 2003 quota document increases in the female share of roughly 30-35 percentage points, alongside evidence of firm restructuring to avoid quota requirements and mixed valuation effects (Ahern and Dittmar 2012; Bertrand et al. 2019; Eckbo, Nygaard, and Thorburn 2022). Within the U.S., research on California's gender quota (SB 826) finds first-stage effects of approximately 10 percentage points in the three years preceding NASDAQ's regulation (Allen and Wahid 2023; Gopal 2025).[7] Some studies find non-negative to positive market responses (Allen and Wahid 2023), while other work reports short-run negative share-price reactions of 1-2% following the quota's adoption or repeal (Hwang, Shivdasani, and Simintzi 2018; Greene, Intintoli, and Kahle 2020; Von Meyerinck et al. 2018; Klick 2025). Institutional investor pressure has also been shown to increase female board representation in magnitudes comparable to quotas (Gormley et al. 2023). In contrast to these quota and pressure-based interventions, we show that mandatory disclosure imposes minimal constraints on firms, as reflected by minimal evasion and smaller increases in diversity induced by the regulation.

[7] As in Gopal (2025), board gender diversity exhibits mean reversion in our sample, motivating the use of difference-in-differences methods rather than the traditional shift-share instrument used in prior literature. We similarly observe a cross-sectional relationship between diversity and firm size in our sample.

Finally, we contribute to the broader literature on the economics of disclosure and financial reporting regulation, as recently reviewed by Leuz and Wysocki (2016). The authors categorize the potential economic benefits of disclosure into seven dimensions: market liquidity, cost of capital, Tobin's Q, investors' portfolio allocations, the analyst and broader information environment, capital raising and structure, and investment behavior. We provide suggestive evidence this disclosure regulation may have increased market liquidity and changed the broader informational environment, potentially contributing to the positive valuation effects. We typically find small or null effects in the other mechanisms we examine – including changes to operating performance, return on assets, and corporate governance actions – so these mechanisms do not appear to explain the positive announcement returns.[8] Leuz and Wysocki (2016) further highlight that a central challenge in this literature is the difficulty of establishing credible counterfactuals – specifically, unaffected control groups and natural experiments that enable clean identification of regulatory effects and their economic consequences. We view our study as making progress in addressing this challenge by using an unaffected control group of NYSE-listed firms and conducting an event study around both adoption and repeal.

2 Legal Context

On December 1, 2020, Nasdaq proposed listing rules on board diversity, which the SEC approved on August 6, 2021 (U.S. Securities and Exchange Commission 2021). Rule 5606 requires every listed company to publish a standardized Board Diversity Matrix reporting the number of directors by gender and demographic background (race or ethnicity and LGBTQ+ status), based on voluntary self-identification. Firms post the matrix annually in the proxy statement or on the company website (Nasdaq 2023). Rule 5605(f) sets a comply-or-explain objective: have at least two diverse directors—one who self-identifies as female and one who self-identifies as an underrepresented

[8] We observe negative effects for Tobin's Q, though its use as a measure of firm value is contested (Bartlett and Partnoy 2020).

minority or LGBTQ+—or an explanation of why the firm falls short.[9] After a December 2022 amendment, firms had to publish a matrix and have (or explain the absence of) one diverse director by December 31, 2023; the two-director objective applied by December 31, 2025 for the Nasdaq Global Select and Global Markets and by December 31, 2026 for the Capital Market (U.S. Securities and Exchange Commission 2022). The rule was challenged in the Fifth Circuit, and on December 11, 2024, the en banc court vacated the SEC's approval order by a 9–8 vote, holding that the SEC had not properly found the rule to be related to the purposes of the Exchange Act (United States Court of Appeals for the Fifth Circuit 2024). Table A1 reports the full timeline. Our analysis focuses on gender, which is better measured than race, ethnicity, or sexual orientation.

2.1 Information Available Prior to the Regulation

Before 2010, boards' gender composition was disclosed mainly through voluntary channels such as corporate social responsibility reports and company websites (Liang et al. 2025). Since 2010, Regulation S-K has required firms to state whether and how their nominating committees consider diversity, but the SEC left diversity undefined and the disclosures contained no director counts (U.S. Securities and Exchange Commission 2009). Investors could often infer directors' gender from names, biographies, and photographs in proxy statements, and could buy data from vendors such as BoardEx and Equilar (Wharton Research Data Services 2026; Equilar 2017). Beginning in 2017, large index investors, starting with State Street's "Fearless Girl" campaign, pressed boards without women, and ISS and Glass Lewis adopted voting policies against nominating committee chairs of all-male boards (State Street Global Advisors 2017; Cahill Gordon & Reindel LLP 2018). Gormley et al. (2023) show that these campaigns led firms to add at least 2.5 times as many female directors in 2019 as in 2016.

[9] Boards of five or fewer directors need one diverse director, and smaller reporting companies and foreign issuers may satisfy the objective with two female directors (U.S. Securities and Exchange Commission 2021).

2.2 What the Rule Added

Investors could often infer gender before the rule but had much less information on race, ethnicity, and sexual orientation, and the information they had varied in coverage and definitions across firms. The rule added standardized, self-identified statistics for every listed firm. The explain option added information that board composition alone cannot reveal, such as how committed a firm is to adding diverse directors, what barriers it faces, and how it plans to overcome them.

3 Data Description and Summary Statistics

3.1 Sample Construction

We construct a firm-year panel from BoardEx, Compustat, and CRSP, linked with the crosswalk provided by WRDS. Table 1 reports the sample size by exchange. The panel covers 2017–2024, but the regressions use 2017–2023 because 2024 is incomplete for some variables. We keep domestic firms listed on NASDAQ (treated) or NYSE (control). In the event studies, we use CRSP ordinary common shares (share codes 10 and 11) with exchange code 1 (NYSE) or 3 (NASDAQ), which excludes ADRs, funds, and NYSE American firms. The panel is unbalanced: we do not drop IPOs, SPACs, or firms that exit, and Table A2 reports IPO and attrition rates by exchange. The synthetic DiD estimates use balanced panels. Table A3 shows the share of BoardEx firm-years matched to CRSP and Compustat, which exceeds 86 percent in every year.

Board composition comes from BoardEx organizational summary files, which list directors as of the company's annual report date. We define the female share as one minus BoardEx's gender ratio (the male share of directors), 1(AMB) as an indicator for a board with no female director, and board size as the number of directors. 1(Expand) indicates a larger board than in the prior year, and 1(Male Dropped) indicates an unchanged board size with a lower male share. We also use BoardEx data on directors' education (MBA, doctorate, Ivy League), tenure, and board, C-suite, and sector experience. Because the roster is dated at the annual report date, an observation reflects

the board at that date rather than at fiscal year-end.

Firm characteristics come from Compustat Fundamentals Annual, matched to BoardEx by year. Return on assets is net income over total assets, leverage is long- and short-term debt over total assets, and cash-to-assets is cash and short-term investments over total assets. Tobin's Q is the market value of equity (common shares outstanding times price) plus long- and short-term debt, divided by total assets. We also use sales, employees, and firm age, measured as years since the Compustat IPO date. The financial outcomes are winsorized at the 1st and 99th percentiles within each year or log-transformed. Industry is the 11 GICS sectors from Compustat, and we classify firms as business-to-business or business-to-consumer following Delgado and Mills (2020). Company policies (mergers, dividends, and changes of at least 5% in shares outstanding) and bid-ask spreads, our measure of liquidity, come from CRSP. Table A4 lists all variables and sources.

To study CEO turnover and compensation, we link ExecuComp to Compustat. We also merge two ESG measures: Sustainalytics risk scores, where 0–10 is negligible risk and 40 or more is severe, and MSCI scores from 0 to 10, where higher scores indicate ESG "leaders." Finally, we measure institutional ownership by the combined stake of the Big Three (BlackRock, Vanguard, and State Street) from 13F filings. We use Big Three ownership as a proxy for how visible a firm's board diversity was to investors before the rule, and we control for it in the board-composition and event-study regressions.

3.2 Sample Description

Table 2 reports 2020 means by exchange, and Table A5 further restricts to firms with all-male boards. NASDAQ boards are smaller and less female than NYSE boards: the average female share is 19% against 25%, and 16% of NASDAQ boards are all-male against 4% of NYSE boards. NASDAQ firms are also smaller, younger, less profitable, less levered, more cash-rich, and have higher Tobin's Q and wider bid-ask spreads. Industry composition differs: Health Care, Information Technology, and Financials are overrepresented on NASDAQ, and Industrials, Consumer Discretionary, Energy, Materials, and Utilities on NYSE. Most of these differences are significant

at the 1% level. Among all-male boards, NASDAQ firms remain smaller and more cash-rich, but board characteristics are largely similar across exchanges.

These level differences matter for identification only if they predict different changes in board composition after 2020. Two features of the data bear on this concept. First, all-male boards were disappearing faster on NASDAQ (Table 1), so we cannot assume parallel pre-trends for all-male boards. Second, NASDAQ's sample grows sharply in 2021, with an IPO wave larger than NYSE's (Table A2), so new listings can shift the composition of the treated group. We therefore include industry-by-year fixed effects in our baseline analysis and also estimate synthetic DiD, which reweights NYSE firms to match NASDAQ's pre-period path on a balanced panel. Finally, firms that did not meet the diversity objective were expected to explain why. We hand-collect these explanations from proxy statements and annual reports, match the firms to CRSP by PERMNO to conduct an event study, and classify the explanations into supply and demand side factors.

4 Results

4.1 Compliance

Unlike Norway's 2003 gender quota, where only about one-third of treated firms remained listed within five years of announcement (Bertrand et al. 2019), we find no unusual pattern of NASDAQ attrition or IPO avoidance around the disclosure rule (Table A2). The rule is soft-touch: reputational pressure and an explain option, and not a hard quota that would push firms off the exchange. Minimal evasion fits the cost-benefit bound in Gopal (2025), under which the upper-bound cost to shareholders is the minimum of the evasion cost and the present value of remaining non-diverse.

Female board share rose on both exchanges over 2017–2023, with NYSE firms consistently higher in the pre-period and a visibly sharper rise on NASDAQ after 2021 that narrowed the gap (Figure 1). These descriptive patterns motivate the difference-in-differences analysis that follows. We compare NASDAQ-listed firms (treated) to NYSE-listed firms (control) with OLS difference-

in-differences and event-study specifications:

$$Y_{fti} = \beta_0 + \beta_1 \big(\text{NASDAQ} \times \mathbb{I}(t > 2020)\big) + \delta_f + \gamma_{ti} + \varepsilon_{fti}, \tag{1}$$

$$Y_{fti} = \theta_0 + \sum_{t \neq 2020} \theta^t \big(1[\text{Year} = t] \times \text{NASDAQ}\big) + \delta_f + \gamma_{ti} + \varepsilon_{fti}, \tag{2}$$

where Y_{fti} is a board-composition outcome for firm f in year t and industry i, δ_f are firm fixed effects, and γ_{ti} are industry-by-year fixed effects. The regressions use an unbalanced firm-year panel for 2017–2023, with standard errors clustered at the firm. Firm fixed effects absorb time-invariant differences across firms. Industry-by-year effects absorb shocks common to an industry in a given year and allow different industry trends—important here because NASDAQ and NYSE differ sharply in sector mix.

Because key outcomes—especially the all-male-board indicator—lack clean parallel pre-trends, we also estimate the synthetic difference-in-differences (SDiD) estimator of Arkhangelsky et al. (2021), which reweights NYSE control firms to match NASDAQ's pre-treatment outcome trajectory.[10] The estimator solves

$$\big(\hat{\beta}^{sdid}, \hat{\mu}, \hat{\delta}, \hat{\gamma}\big) = \arg\min_{\beta,\mu,\delta,\gamma} \left\{ \sum_{f=1}^{N} \sum_{t=2017}^{2023} \big(Y_{ft} - \mu - \delta_f - \gamma_t - D_{ft}\beta^{sdid}\big)^2 \hat{\omega}_f^{sdid} \hat{\lambda}_t^{sdid} \right\}, \tag{3}$$

where $D_{ft} = 1$ for NASDAQ firms after 2020, and the unit and time weights $\hat{\omega}_f^{sdid}$ and $\hat{\lambda}_t^{sdid}$ are chosen to minimize pre-period trend differences. Inference uses jackknife standard errors clustered at the firm. Jackknife is well suited here because the donor pool is smaller than the treated group and the estimate can be sensitive to units that receive large weight; the bootstrap is infeasible when control units are fewer than treated units.

In the baseline DiD, mandatory disclosure raises the female board share by 0.8 percentage points, about 4% of a 20% mean (Table 3, Panel A, column 1). Adjustment is concentrated on the

[10] Relative to classical synthetic control, SDiD targets pre-treatment *trends* rather than levels, remains invariant to additive unit shocks, and derives both unit and time weights from the outcome data alone, limiting researcher discretion (Abadie 2021; Ferman and Pinto 2021; Arkhangelsky et al. 2021).

extensive margin: the share of all-male boards falls by 8.9 percentage points (Table 3, Panel A, column 2). That pattern matches the rule's design: firms were first required to have at least one diverse director or explain why they did not, and boards of five or fewer needed only one, so the most direct response was for all-male boards to add a single woman. We do not, however, claim clean identification for the all-male-board outcome: its event-study coefficients already decline through 2017–2019, so part of the pooled DiD may reflect ongoing convergence rather than the rule alone. As a check, restricting the sample to firms that were already gender-diverse before the rule yields near-zero and insignificant post-period event-study coefficients for both male share and all-male boards (Table A6), which reinforces that the first-order response is among boards that entered treatment all-male.[11]

The SDiD estimator produces qualitatively similar results: a 4.5 percentage-point decline in all-male boards (Table A7, column 2). Effects on board size, expanding the board, and dropping a male director while holding size fixed are at most marginally significant (Table 3, Panel A, columns 3–5), consistent with a small intensive-margin shift in female share. To probe mechanisms, we next examine heterogeneity informative about reputational pressure. If disclosure raises the cost of maintaining a non-diverse board, responses should be stronger where outside scrutiny is higher. Two patterns fit that channel. First, the estimated decline in the male share of directors is larger among firms with above-median pre-rule ESG ratings (Figure 2). Second, the baseline estimates are essentially unchanged when we control for firm size with log revenue, so the baseline effect is not a pure size-composition story. Across cuts, the decline in all-male boards remains the dominant margin.

If the mandate works through disclosure and reputation, we expect targeted board adjustments rather than broad governance overhauls. Consistent with that view, we find no statistically meaningful effect on CEO turnover, and CEO compensation is largely unchanged aside from a B2B subsample estimate that we interpret cautiously. Board credentials move selectively: MBA and Ivy League shares rise in several specifications, while most board, C-suite, and sector experience

[11] Several pre-period interactions remain nonzero in that restricted sample, so the exchanges differ systematically among already-diverse firms; pooled post-period DiD estimates should therefore be read cautiously.

measures remain near zero—suggesting modest rebalancing rather than a wholesale reshaping of boards.

4.2 Effect on Medium-Run Financial Outcomes

We next ask whether the modest first-stage change in board gender composition (Section 4.1) affects firms' medium-run financial outcomes. Given a 0.8 percentage-point rise in female share over 2021–2023, large changes in operating performance are unlikely. More plausible margins are market-based valuations and the information environment. We report difference-in-differences and event-study estimates for return on assets, return on equity, cash flow to assets, Tobin's Q, market-to-book, and relative bid-ask spreads (Table A8), and synthetic difference-in-differences estimates that reweight the comparison group and time periods to better match pre-treatment outcomes on a balanced panel (Tables A9 and A10).

Liquidity is the one financial margin that moves in the direction an information-asymmetry channel would predict. Relative bid-ask spreads fall by 0.002 in the pooled DiD, against a mean of 0.006, and by 0.0009 under SDiD (Tables A8 and A9). The compression is concentrated in 2021–2022. In the OLS event study it reverses in 2023 (+0.006), and under SDiD the 2023 coefficient is a small and insignificant increase (Tables A8 and A10). A lower pooled spread is consistent with improved liquidity after standardized disclosure, but the 2023 reversal means we cannot claim a lasting change in the information environment.

In the pooled DiD, operating performance measures – return on assets, return on equity, and cash flow to assets – are all small and statistically indistinguishable from zero (Table A8, Panel A). The SDiD estimates on the balanced panel are likewise close to zero and insignificant (Table A9, Panel A), as is ordinary DiD on that same balanced sample (Table A9, Panel B). The event studies are mixed rather than simply uninformative. Several pre-period differences are significant: return on assets and cash flow to assets are lower on NASDAQ in 2017–2019 (Table A8, Panel B). Some 2021 estimates look favorable—return on assets rises by 0.012 in the OLS event study and by 0.026 under SDiD—but those coefficients turn negative by 2022–2023 (Tables A8 and A10).

We therefore read the event studies as showing no sustained post-period improvement against a backdrop of pre-existing differences. The same ROA null appears across the heterogeneity cuts in Table 4: high-ESG firms, larger boards, male-dominated industries, and B2B firms.

Interpreting the market-based ratios requires more caution, because Tobin's Q and market-to-book load directly on price. The pooled DiD associates the rule with a 0.217 decline in Tobin's Q and a 0.103 decline in log market-to-book, both significant at the 1% level (Table A8, Panel A). SDiD, which uses log Q, delivers the same qualitative pattern: -0.107 for log Q and -0.134 for log market-to-book (Table A9, Panel A). Ordinary DiD on the balanced panel is smaller but still negative and significant. Two caveats keep us from treating this as a clean valuation loss. First, the OLS event study for Q already trends down in 2017–2019, and SDiD shrinks but does not eliminate those pre-period gaps (Tables A8 and A10). Second, a medium-run decline in Q sits uncomfortably next to the positive announcement returns around SEC approval. The decline in log Q is also fairly uniform across the Table 4 cuts: it remains negative and significant for high-ESG firms (-0.058), with a size control (-0.068), on larger boards (-0.067), in male-dominated industries (-0.097), and among B2B firms (-0.083); the triple-difference estimate is similar in magnitude but imprecise. We therefore treat the Q and market-to-book estimates as suggestive of lower subsequent valuations, not as a well-identified effect of the rule on fundamentals.

Finally, we ask whether firms induced to add more women saw different financial outcomes, following the instrumental-variables specification in Ahern and Dittmar (2012). We instrument a firm's current female board share with interactions between post-rule year indicators and its 2020 female share, separately for NASDAQ and NYSE (Table A11). Firms that started with fewer women had more room to adjust, so those interactions should predict subsequent increases in female share—and more so on NASDAQ if the rule was binding. The first stage is strong on both exchanges and somewhat stronger on NASDAQ ($F = 169$ versus 103). The second stage, however, does not reveal a stable link from board gender composition to firms' medium-run financial outcomes. Industry-adjusted Tobin's Q, return on assets, market-to-book, and cash flow to assets are all insignificant. The only significant coefficient is for industry-adjusted return on equity, and

the exchanges disagree: it is negative and significant at the 1% level on NASDAQ and positive and significant at the 10% level on NYSE. Taken with the reduced-form evidence, we find no robust evidence of medium-run effects of the disclosure-induced change in board composition on firm performance.

4.3 Market Reaction

We next examine how investors responded to two regulatory events: the SEC's approval of NASDAQ's board-diversity rule on August 6, 2021, and the Fifth Circuit's en banc vacatur of that approval on December 11, 2024 (a 9–8 decision). We use a two-day window—the event date and the next trading day—because news released late in the session may not be fully incorporated by the close.

Following Greene, Intintoli, and Kahle (2020), we report firm-level abnormal returns and a calendar-time portfolio test in the spirit of Jaffe (1974). Baseline abnormal returns come from a four-factor model that adds momentum to the Fama–French factors:

$$AR_{f,t} = R_{f,t} - \left(\hat{\beta}_{f,0} + \hat{\beta}_{f,M} R_{M,t} + \hat{\beta}_{f,SMB} SMB_t + \hat{\beta}_{f,HML} HML_t + \hat{\beta}_{f,MOM} MOM_t\right). \tag{4}$$

$R_{f,t}$ is the firm's return, $R_{M,t}$ is the value-weighted excess return on CRSP firms incorporated in the United States, and SMB_t, HML_t, and MOM_t are the daily size, value, and momentum factors from Kenneth French's data library. Cumulative abnormal returns sum these residuals over the two-day window. Factor loadings are estimated over 252 trading days, with a 30-day gap before the event; we require at least 100 return observations in the estimation window.

Because the same news hits many firms on the same day, returns are contemporaneously correlated. We therefore report a standardized cross-sectional t-test, Patell's Z, a conventional cross-sectional t-test, and two nonparametric tests (the generalized sign test and the Wilcoxon signed-rank test). We also adopt the portfolio approach, which addresses the same dependence by collapsing firms into a single series.

The average two-day CAR around approval is 0.748% for NASDAQ firms and essentially zero for NYSE firms (Table 5). The NASDAQ reaction is larger among firms that entered the event with all-male boards (1.483%). These estimates are consistent with investors viewing approval as value-increasing, especially at previously all-male firms. One limit on that reading should be kept in view: the calendar-time portfolio test below does not detect a significant approval effect, so the firm-level CARs for adoption are not our strongest result.

The vacatur on December 11, 2024, moves prices in the opposite direction. NASDAQ firms post a two-day CAR of -1.307%, and the decline is again larger among all-male-board firms (Table 6). NYSE firms are not a complete zero: they fall by 0.202% on the event day itself, significant at the 1% level under each of the reported tests, but are flat the next day, for a two-day CAR of -0.175%. The NASDAQ two-day move is several times larger. The significant NYSE decline on December 11 nevertheless means we cannot treat that window as free of market-wide news.

We also split NASDAQ firms by whether they had filed an explanation for falling short of the diversity objectives (Table A12). Explaining is a permitted response under a comply-or-explain rule, so this split is not a comparison of firms that "complied" with firms that did not. Non-explaining firms ($N = 2{,}075$) have a two-day CAR of -1.320%. The 41 explaining firms have a two-day CAR of -0.640% that is statistically insignificant. We do not read that contrast as evidence that explaining firms were unaffected, as the subsample is too small for a sharp test.

As a check on whether the event-day CARs are unusual relative to other days, we assign placebo event dates to every trading day between 252 and 30 days before each actual event (e.g. Cohn et al. 2026), using the same NASDAQ sample and the same $(0, 1)$ window. Figure 3 shows the resulting distributions. The approval CAR exceeds 207 of the placebo CARs (92%), with only 16 placebos larger. The vacatur CAR lies in the left tail: only two placebos (0.92%) are more negative (Figure 3, panels (a) and (c)). The event-day returns are therefore unusual relative to other days in the estimation window.

We also compare NASDAQ and NYSE two-day CARs in a multivariate regression with in-

dustry fixed effects (e.g. Ahern and Dittmar 2012), reported in Tables 7 and 8. In the baseline specification, the NASDAQ coefficient is 0.506 percentage points around approval and -0.519 around the vacatur (the latter significant at the 10% level). The NASDAQ dummy is not significant in every column: it loses significance once the NASDAQ–diversity interactions are included, and in some of the specifications that add Big Three ownership. Adding an indicator for any woman director shrinks the NASDAQ coefficient on both dates. That indicator itself is -0.899 around approval and 0.963 around the vacatur, so already-diverse firms have more muted returns in both windows. The NASDAQ $\times$ diversity interactions are imprecise, however, so the multivariate tables should not be read as isolating a well-estimated NASDAQ-specific diversity slope.

The same muted pattern appears for Big Three ownership, which we use as a proxy for prior investor attention to board gender diversity (Gormley et al. 2023). Greater Big Three ownership is associated with less positive CARs around approval and less negative CARs around the vacatur, and the NASDAQ $\times$ Big Three interaction goes in the same direction. These estimates are consistent with more muted share-price reactions when information about gender diversity was already known to the market. Beginning in 2017, the Big Three's campaigns and voting policies drew attention to gender diversity at the firms they held most heavily, so investors in those firms were more likely to have already priced diversity-related information. A complementary interpretation is that high-Big-Three firms already have more female directors—the correlation is 0.34—and so had less scope to be affected by the rule.

4.3.1 Robustness Test: Portfolio Approach

As a robustness test, we form an equal-weighted calendar-time portfolio of NASDAQ-listed firms, following Jaffe (1974) and as recently applied by Allen and Wahid (2023) and Eckbo, Nygaard, and Thorburn (2022). We estimate

$$R_{p,t} = \beta_0 + \beta_1 R_{M,t} + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 MOM_t + AR \times D_t + \varepsilon_t, \tag{5}$$

where $R_{p,t}$ is the portfolio's average daily return minus the one-month Treasury bill rate, and D_t equals one on the event day and the next trading day. The coefficient AR is the portfolio's average daily abnormal return during the window. The remaining factors are as in equation (4). The estimation sample runs from 365 calendar days before the event through the day after it. Because firms are collapsed into a single return series, this test is robust to the cross-sectional correlation in returns that arises when all firms face the same event date.

Around approval, AR is 0.004 for all NASDAQ firms and 0.010 for NASDAQ all-male-board firms, and neither is statistically significant (Table 9). The portfolio test therefore does not confirm the positive firm-level approval CARs reported above. One possible reason is that approval was partly anticipated: NASDAQ filed the proposal in December 2020, and the SEC's approval followed months of public comment.

Around the vacatur, AR is negative and significant: -0.006 for all NASDAQ firms and -0.013 for NASDAQ all-male-board firms, or roughly -1.2 and -2.6 percentage points over the two-day window (Table 10). The corresponding NYSE estimates are small and insignificant. The narrow 9–8 en banc vote is consistent with the outcome being a genuine surprise.

We also run portfolio-level placebo tests, assigning pseudo-event dates to every trading day between 252 and 30 trading days before each actual event and re-estimating equation (5) with D_t redefined accordingly (Figure 4). The approval coefficient exceeds 195 of the placebo coefficients (87%), with 20 larger, so it is high in the distribution but not extreme, consistent with its insignificant point estimate (panel (a)). The vacatur coefficient lies deep in the left tail: only four placebos (1.8%) are more negative (panel (c)).

Finally, the same regression around the other NASDAQ-rule dates in Table A1, excluding approval and vacatur, produces insignificant coefficients on D_t (Table 11). Taken together, the most robust share-price evidence is the negative NASDAQ reaction to the vacatur, which is larger among all-male-board firms and absent among NYSE firms. The approval evidence points in the same direction, with investors appearing to value the rule, but it holds in the firm-level tests and not in the portfolio test.

4.4 Textual Analysis of Explanations

The rule's comply-or-explain structure gives a rare window into why firms do not diversify their boards. Firms that chose to explain rather than add diverse directors had to state their reasons publicly, so these disclosures are direct, firm-authored evidence on the barriers to board diversity that board-composition data alone cannot reveal. Forty-five firms provided such explanations, and we classify them into three categories: supply-side constraints, demand-side constraints, and other reasons. Supply-side constraints concern the availability or qualifications of candidates, such as a limited pipeline or an emphasis on merit-based selection. Demand-side constraints concern the cost of adding diverse directors or restructuring the current board. Other reasons include statements that the current board is already optimal and commitments to diversify later.

Supply-side constraints are the most common, accounting for 51.1% of explanations (Figure 5). The largest group, 31.1% of all explanations, is merit-based. For example: "Rather than considering the level of representation of female and/or demographically diverse individuals for director and executive officer positions when making Board of Directors or executive officer appointments, the Company considers all candidates based on their merit and qualifications relevant to the specific role and their experience as a director of our company over the past several years." The remaining 20.0% cite a limited candidate pipeline. For example: "As of November 29, 2024, our Board has not identified potential nominees having an interest in serving on the Board that qualify as Diverse, while possessing the skillsets and other qualifications to appropriately represent the interests of shareholders and provide strategic oversight of our business."

Demand-side constraints account for 17.8% of explanations and mostly cite the cost of expanding or restructuring the board. The remaining 31.1% fall into the other category. These are firms arguing that the current board composition is already optimal (17.8%), typically because replacing a director solely to increase diversity would sacrifice existing expertise, and firms that acknowledge the value of diversity but defer changes to a later date (13.3%).

5 Conclusion

We examine how mandatory disclosure of board diversity affects firm behavior. The setting is NASDAQ's board diversity rule, which required listed firms to disclose board composition and either have at least two diverse directors, one of whom is female, or explain why they did not. We find that the rule led to a modest increase in female board representation among NASDAQ-listed firms. The increase is smaller than those associated with institutional investors' diversity campaigns (Gormley et al. 2023), gender quota mandates (Gopal 2025), and mandatory diversity disclosure policies in other countries (Bakke et al. 2021; Hu, Hung, and Li 2025). The modest response is consistent with firms perceiving little reputational cost from explaining rather than complying, and suggests that in the U.S. context, a disclosure-based approach alone may not be sufficient to drive substantial progress in board gender diversity.

We also study investors' response to the rule. NASDAQ firms experienced negative abnormal returns when the Fifth Circuit vacated the rule, a result that holds across firm-level and portfolio tests and is larger among firms with all-male boards. Firm-level returns around the SEC's approval were positive, although the portfolio test does not confirm them. Together, these results suggest that investors valued the disclosure framework. Consistent with the rule reducing information frictions, bid–ask spreads narrowed after the rule took effect, and announcement returns were more muted at firms with greater Big Three ownership, where information about board diversity was more widely available beforehand. Over the medium run, however, we find no robust improvement in operating performance and a decline in Tobin's Q that is sensitive to pre-existing trends. Finally, among firms that explained rather than complied, most cite supply-side barriers, especially a commitment to merit-based selection and a limited pipeline of qualified candidates, as the main reasons for not meeting the rule's diversity objectives.

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Table 1: Sample Size

	All firms	NASDAQ			NYSE		
Year		N	N: AMB	Pr(AMB)	N	N: AMB	Pr(AMB)
2017	3,205	1,996	725	0.36	1,209	179	0.15
2018	3,215	2,019	604	0.30	1,196	128	0.11
2019	3,222	2,044	418	0.20	1,178	84	0.07
2020	3,319	2,157	349	0.16	1,162	44	0.04
2021	3,971	2,638	344	0.13	1,333	56	0.04
2022	3,844	2,554	280	0.11	1,290	44	0.03
2023	3,545	2,311	237	0.10	1,234	29	0.02
2024	2,948	1,797	115	0.06	1,151	16	0.01

Notes: The sample restricts to companies that report board gender. The annual gender composition of corporate boards is provided by BoardEx and reflects the board's composition as of the company's annual report date. We also restrict to companies matched between BoardEx, Compustat, and CRSP using the crosswalk provided by WRDS. AMB refers to companies with All-Male Boards. NASDAQ's gender disclosure requirement required that all firms listed on NASDAQ either have at least one woman or minority represented on their corporate board or provide a reason. This rule was approved on August 6th, 2021.

Table 2: Summary Statistics in 2020

	Mean				Observations	
	NASDAQ	NYSE	Diff.	p-value	NASDAQ	NYSE
Board Characteristics						
Board Size	8.03	9.57	1.54***	0.000	2,157	1,162
% Directors with MBA	0.34	0.41	0.07***	0.000	1,909	1,107
% Directors with Doctoral Degree	0.11	0.06	−0.05***	0.000	1,909	1,107
% Directors with Ivy League Degree	0.25	0.28	0.03***	0.000	1,909	1,107
Average Director Tenure	0.05	0.05	−0.00	0.446	1,911	1,107
% Directors with Board Exp	0.81	0.84	0.02***	0.003	1,911	1,107
% Directors with C-Suite Exp	0.68	0.72	0.04***	0.000	1,911	1,107
% Directors with Sector Exp	0.66	0.48	−0.18***	0.000	1,911	1,107
1(AMB)	0.16	0.04	−0.12***	0.000	2,157	1,162
1(Expand)	0.23	0.27	0.04***	0.007	2,157	1,162
Age	13.30	16.07	2.77***	0.000	1,310	560
Female Ratio	0.19	0.25	0.06***	0.000	2,157	1,162
Number of Employees (thousands)	5.24	23.85	18.60***	0.000	2,049	1,128
Business-to-Business	0.64	0.66	0.02	0.435	1,732	814
Business-to-Consumer	0.36	0.34	−0.02	0.435	1,732	814
Firm Characteristics						
Return on Assets	−0.05	0.00	0.05***	0.000	2,076	1,137
Log(Tobin's Q)	0.39	0.23	−0.16***	0.000	2,020	1,090
Total Assets ($ mill)	4,813.97	34,254.94	29,440.97***	0.000	2,077	1,137
Sales ($ mill)	521.18	2,579.58	2,058.40***	0.000	2,076	1,137
Leverage	0.23	0.35	0.12***	0.000	2,037	1,098
Cash to Assets	0.35	0.15	−0.20***	0.000	2,077	1,137
Bid-Ask Spread	0.01	0.00	−0.01***	0.000	2,158	1,161
Company Policy						
1(Merger)	0.00	0.00	−0.00	0.951	2,157	1,162
1(Dividend)	0.28	0.60	0.31***	0.000	2,157	1,162
1(Dec in Shares Outstanding $\geq$ 5%)	0.09	0.06	−0.03***	0.002	2,157	1,162
1(Incr in Shares Outstanding $\geq$ 5%)	0.06	0.02	−0.04***	0.000	2,157	1,162
ESG						
Sustainalytics ESG Score	29.52	27.49	−2.03***	0.000	1,142	941
MSCI ESG Score	4.50	4.69	0.19***	0.000	931	950
1(ESG High)	0.42	0.55	0.13***	0.000	930	946
Industry						
Energy	0.02	0.08	0.06***	0.000	2,157	1,162
Materials	0.01	0.09	0.07***	0.000	2,157	1,162
Industrials	0.10	0.20	0.11***	0.000	2,157	1,162
Consumer Discretionary	0.09	0.15	0.06***	0.000	2,157	1,162
Consumer Staples	0.03	0.04	0.01*	0.057	2,157	1,162
Health Care	0.34	0.08	−0.26***	0.000	2,157	1,162
Financials	0.20	0.15	−0.04***	0.002	2,157	1,162
Information Technology	0.15	0.08	−0.06***	0.000	2,157	1,162
Communication Services	0.04	0.04	−0.00	0.860	2,157	1,162
Utilities	0.01	0.05	0.04***	0.000	2,157	1,162
Real Estate	0.01	0.02	0.01**	0.025	2,157	1,162

Continued on next page

(Table 2 continued)

	Mean				Observations	
	NASDAQ	NYSE	Diff.	*p*-value	NASDAQ	NYSE

Notes: The sample restricts to an unbalanced panel of firms that were domestic and listed in 2020. The table comes from a cross section of the data, representing 2020 values. Raw means and p-values from a two-sided t-test are reported. Boardroom characteristics are derived from BoardEx and represent mean values in 2020. Firm characteristics are derived from Compustat and represent mean values in 2020. Return on Assets is net income before extraordinary items and discontinued operations divided by book assets. Tobin's Q is the ratio of the firm's market value to its book value of assets. Total Assets refers to Compustat item AT. Sales refers to Compustat item SALE. Leverage refers to book value of long- and short-term debt divided by total assets. Cash to Assets refers to cash and short-term investments divided by total assets. All company policies indicate if the event occurred for some security during the calendar year, and are derived from CRSP's Events files. Codes for industry classification are also derived from Compustat.

Table 3: Effect of Mandatory Disclosure on Board Composition

Dependent variable:	Male Share of Board (1)	1(All-Male Board) (2)	Board Size (3)	1(Expand Board) (4)	1(Male Dropped) (5)
Panel A: Intent to Treat Effect Estimates					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.008^{***}	-0.089^{***}	-0.074^{*}	0.021^{*}	-0.001
	(0.003)	(0.009)	(0.045)	(0.013)	(0.008)
Panel B: Intent to Treat Effect Estimates with Control for Big Three Ownership Share					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.011^{***}	-0.082^{***}	-0.076	0.024^{*}	0.003
	(0.003)	(0.010)	(0.050)	(0.014)	(0.009)
Big Three ownership	-0.155^{***}	-0.346^{***}	1.494^{***}	-0.007	-0.029
	(0.025)	(0.082)	(0.317)	(0.118)	(0.064)
Panel C: Event Studies Estimates					
NASDAQ $\times$ Year = 2017	0.000	0.082^{***}	0.077	0.010	-0.020
	(0.004)	(0.015)	(0.059)	(0.024)	(0.013)
NASDAQ $\times$ Year = 2018	0.004	0.066^{***}	0.094^{*}	0.035	-0.024^{*}
	(0.004)	(0.013)	(0.053)	(0.025)	(0.014)
NASDAQ $\times$ Year = 2019	0.002	0.017^{*}	0.075^{*}	0.045^{*}	-0.016
	(0.003)	(0.009)	(0.044)	(0.027)	(0.015)
NASDAQ $\times$ Year = 2021	-0.006^{**}	-0.031^{***}	-0.027	0.033	-0.015
	(0.003)	(0.008)	(0.044)	(0.026)	(0.015)
NASDAQ $\times$ Year = 2022	-0.009^{***}	-0.056^{***}	0.009	0.061^{**}	-0.003
	(0.003)	(0.009)	(0.053)	(0.024)	(0.014)
NASDAQ $\times$ Year = 2023	-0.005	-0.065^{***}	-0.027	0.037	-0.028^{**}
	(0.004)	(0.010)	(0.058)	(0.024)	(0.014)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year-SIC FE	Yes	Yes	Yes	Yes	Yes
Observations	25,766	25,766	25,766	23,958	23,958
Dependent variable mean	0.783	0.133	8.63	0.266	0.066
Number of firms	4,593	4,593	4,593	4,356	4,356

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the firm level are reported in parentheses. Panel A reports difference-in-differences specifications with firm and Year-SIC fixed effects. The difference-in-difference treatment effect is a Treated (NASDAQ) $\times$ Post (Year $>$ 2020) interaction. Panel B reports the same difference-in-differences specification controlling for Big Three institutional ownership (measured at the end of 2020). Panel C reports event studies with firm and Year-SIC fixed effects, relative to 2020 baseline. All regressions use an unbalanced panel of firms with available board composition data covering 2017–2023. 'Expand Board' and 'Male Dropped' indicators are available starting in 2018. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. The outcome variables are derived from BoardEx's organizational summary files, which provide the director roster as of the company's annual report date. The 'Expand Board' indicator equals one if board size increases relative to the prior year. 'Male Dropped' equals one if board size is unchanged relative to the prior year while the proportion of male directors fall. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table 4: Effects of Mandatory Disclosure: Heterogeneity

	Baseline	High ESG	Size Control	Large Board	Male Industry	B2B	Triple Diff
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Panel A: Board Composition						
Male Share of Board	−0.008***	−0.018***	−0.008***	−0.012***	−0.004	−0.015***	0.009
	(0.003)	(0.005)	(0.003)	(0.003)	(0.004)	(0.005)	(0.013)
1(All-Male Board)	−0.089***	−0.050***	−0.084***	−0.053***	−0.088***	−0.122***	0.043
	(0.009)	(0.011)	(0.009)	(0.008)	(0.013)	(0.014)	(0.062)
Observations	25,766	6,590	24,011	15,315	14,609	13,500	25,766
	Panel B: Financial Outcomes						
Log(Q)	−0.076***	−0.058***	−0.068***	−0.067***	−0.097***	−0.083***	−0.091
	(0.014)	(0.022)	(0.014)	(0.015)	(0.021)	(0.021)	(0.085)
Observations	22,725	5,967	21,382	13,619	12,785	12,063	22,725
RoA	−0.001	−0.006	−0.005	0.001	0.003	0.002	−0.005
	(0.004)	(0.005)	(0.005)	(0.004)	(0.006)	(0.006)	(0.03)
Observations	25,453	6,575	24,011	15,254	14,424	13,443	25,453
	Panel C: Board Experience						
Share with MBA	0.015**	0.030***	0.014**	0.023***	0.029***	0.007	0.002
	(0.006)	(0.009)	(0.006)	(0.007)	(0.008)	(0.010)	(0.041)
Share with Doctoral Degree	−0.000	−0.005	0.001	−0.003	0.000	−0.001	−0.014
	(0.003)	(0.005)	(0.003)	(0.003)	(0.005)	(0.005)	(0.020)
Share from Ivy League	0.011*	0.023***	0.013**	0.013**	0.008	0.018**	−0.053
	(0.006)	(0.008)	(0.006)	(0.006)	(0.008)	(0.009)	(0.035)
Observations	19,035	6,231	18,020	12,950	11,074	9,516	19,035
Share with Tenure	−0.001	−0.012**	−0.001	−0.005	0.002	−0.004	−0.006
	(0.003)	(0.006)	(0.003)	(0.004)	(0.004)	(0.006)	(0.018)
Share with Board Experience	−0.013**	−0.010	−0.013**	−0.008	−0.018**	−0.016*	0.030
	(0.006)	(0.010)	(0.006)	(0.006)	(0.007)	(0.009)	(0.028)
Share with C-Suite Experience	0.002	0.006	0.002	0.010*	0.001	−0.003	−0.006
	(0.006)	(0.008)	(0.006)	(0.006)	(0.008)	(0.009)	(0.033)
Share with Sector Experience	−0.005	−0.011	−0.006	−0.008	−0.002	−0.005	0.023
	(0.006)	(0.010)	(0.006)	(0.007)	(0.008)	(0.010)	(0.027)
Observations	19,044	6,231	18,028	12,954	11,080	9,519	19,044
	Panel D: CEO Outcomes						
CEO Turnover	0.004	−0.006	0.005	0.007	0.023	−0.018	0.003
	(0.013)	(0.019)	(0.013)	(0.014)	(0.018)	(0.018)	(0.013)
Observations	12,192	5,298	12,165	9,591	5,531	6,216	12,192
Log CEO Compensation	0.007	0.057	−0.011	−0.010	0.065	0.094*	0.165
	(0.040)	(0.047)	(0.040)	(0.047)	(0.041)	(0.052)	(0.327)
Observations	12,114	5,309	12,086	9,560	5,509	6,189	12,114

Notes: Clustered (firm) standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The sample restricts to an unbalanced panel of firms that were domestic and listed covering 2017–2023. The table presents the coefficients and standard errors from the difference-in-differences model, unless otherwise specified. Standard errors are clustered at the firm level. The reported rows are restricted to outcome variables that showed significance in the baseline difference-in-difference model. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. Male share of board members is derived from BoardEx's organizational summary files, which provide the director roster as of the company's annual report date. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. Financial variables are derived from Compustat's annual fundamental files and are either log transformed or winsorized at the 1st and 99th percentiles. The percentiles are calculated relative to all domestic and listed companies observed in the annual distribution. The index of financial outcomes averages the z-score across several financial outcomes, following Kling, Liebman, and Katz (2007). For each financial outcome, the z-score subtracts the mean of the control group, then divides by the standard deviation of the control group. Column 2 subsets to companies with above-median MSCI industry-adjusted ESG ratings. Column 3 adds a control for firm size, which is proxied by Log(Revenues). Column 4 subsets to companies that had more than 8 directors (the median board size) in 2020. Column 5 subsets to firms in industries with below-average female board representation. Industry classification and averages are calculated using the 2020 cross-section. Column 6 subsets to business-to-business (B2B) firms. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table 5: Abnormal Returns on August 6, 2021

	Day relative to event	No. of firms	Mean (%)	Tests of mean = 0			
				Std. cross-sectional t-test	Patell's Z test	Gen. sign test	Wilcoxon signed-rank test
	Panel A: All Firms						
NASDAQ	0	1,885	0.247	***	***	***	***
	1		0.501	***	***	***	***
NYSE	0	1,132	0.057				
	1		−0.051		*	***	***
	Panel B: All-Male Board Firms						
NASDAQ	0	279	0.372		**	***	*
	1		1.111	***	***	***	***
NYSE	0	42	−0.338		*		
	1		1.557	**	**	**	**

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The mean on day 0 represents the average abnormal returns on the event date, while the mean on day 1 reflects the average abnormal returns on one day following the event date. For the abnormal returns, we employed an estimation window of 252 days, included a gap of 30 days between the end of the estimation window and the event date (Aug 06, 2021), and only considered firms with at least 100 stock return observations in the estimation window.

Table 6: Abnormal Returns on December 11, 2024

	Day relative to event	No. of firms	Mean (%)	Tests of mean = 0			
				Std. cross-sectional t-test	Patell's Z test	Gen. sign test	Wilcoxon signed-rank test
	Panel A: All Firms						
NASDAQ	0	2,116	−0.727	***	***	***	***
	1		−0.580	***	***	**	***
NYSE	0	1,193	−0.202	***	***	***	***
	1		0.027				
	Panel B: All-Male Board Firms						
NASDAQ	0	185	−1.117	*	***	***	***
	1		−1.216	***			***
NYSE	0	28	−0.003				
	1		0.063				

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The mean on day 0 represents the average abnormal returns on the event date, while the mean on day 1 reflects the average abnormal returns on one day following the event date. For the abnormal returns, we employed an estimation window of 252 days, included a gap of 30 days between the end of the estimation window and the event date (Dec 11, 2024), and only considered firms with at least 100 stock return observations in the estimation window.

Table 7: OLS Regressions on Abnormal Announcement Returns (%) for the Event Date August 06, 2021

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(NASDAQ)	0.506***	0.402**	−0.062	0.428**	0.607	0.132	0.721*
	(0.195)	(0.197)	(0.831)	(0.198)	(0.418)	(0.202)	(0.408)
Women directors > 0		−0.899***	−1.324*				
		(0.282)	(0.791)				
NASDAQ $\times$ Women directors > 0			0.486				
			(0.846)				
Percentage women directors				−0.015**	−0.009		
				(0.007)	(0.013)		
NASDAQ $\times$ Percentage women directors					−0.008		
					(0.016)		
Big Three ownership						−6.270***	−3.849**
						(0.971)	(1.752)
NASDAQ $\times$ Big Three ownership							−3.483*
							(2.098)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.019	0.023	0.023	0.021	0.021	0.033	0.034
Observations	2,880	2,880	2,880	2,880	2,880	2,880	2,880

Notes: ***, ** and * denote statistical significance at the 1%, 5% and 10% levels, respectively. The sample consists of firms listed on the NASDAQ and NYSE. The dependent variable is the two-day cumulative abnormal return (CAR), expressed as a percentage, with day 0 corresponding to August 6, 2021. NASDAQ is a dummy variable equal to one if the firm is listed on NASDAQ. Women directors > 0 is a dummy variable equal to one if the firm has at least one female director. Percentage women directors is the percentage of women on the firm's board. Standard errors are heteroskedasticity-robust.

Table 8: OLS Regressions on Abnormal Announcement Returns (%) for the Event Date December 11, 2024

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1(NASDAQ)	−0.519*	−0.454*	−1.283	−0.495*	−0.728	−0.219	−1.249**
	(0.269)	(0.270)	(1.407)	(0.272)	(0.669)	(0.283)	(0.568)
Women directors > 0		0.963**	0.223				
		(0.488)	(1.325)				
NASDAQ $\times$ Women directors > 0			0.855				
			(1.425)				
Percentage women directors				0.005	−0.001		
				(0.009)	(0.018)		
NASDAQ $\times$ Percentage women directors					0.008		
					(0.021)		
Big Three ownership						4.220***	0.186
						(1.273)	(2.313)
NASDAQ $\times$ Big Three ownership							5.772**
							(2.763)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.043	0.044	0.044	0.043	0.043	0.046	0.047
Observations	3,253	3,253	3,253	3,253	3,253	3,253	3,253

Notes: ***, **, * indicates statistical significance at the 1%, 5%, and 10% levels, respectively. The sample consists of firms listed on the NASDAQ and NYSE. The dependent variable is the two-day CAR (Cumulative Abnormal Return) expressed in percentage, with day 0 corresponding to Dec 11, 2024. The NASDAQ variable is a dummy variable equal to 1 if the firm is listed on NASDAQ. The women directors > 0 variable is a dummy variable equal to 1 if the firm has at least one female director. The percentage of women directors variable represents the percentage of women on the firm's board. Standard errors are heteroskedasticity-robust.

Table 9: Market Reaction Analysis Using Portfolio Approach: Evidence from August 6, 2021

	(1) NYSE AMB	(2) NYSE	(3) NASDAQ AMB	(4) NASDAQ
Intercept	0.002^{***}	0.000	0.001^{*}	0.000
	(0.001)	(0.000)	(0.001)	(0.000)
MKT	0.865^{***}	1.015^{***}	0.716^{***}	0.849^{***}
	(0.062)	(0.016)	(0.067)	(0.032)
SMB	1.004^{***}	0.506^{***}	1.155^{***}	0.950^{***}
	(0.074)	(0.020)	(0.081)	(0.039)
HML	0.257^{***}	0.453^{***}	-0.137^{**}	-0.026
	(0.063)	(0.017)	(0.069)	(0.032)
MOM	-0.087^{*}	-0.093^{***}	0.039	-0.058^{**}
	(0.049)	(0.013)	(0.053)	(0.025)
AR	0.007	-0.001	0.010	0.004
	(0.006)	(0.002)	(0.007)	(0.003)
Observations	254	254	254	254
Adjusted R-squared	0.711	0.968	0.668	0.885

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table presents an analysis of stock returns based on daily portfolios, following the methodology of Eckbo, Nygaard, and Thorburn (2022). The dependent variable is the daily equally weighted average returns in excess of the daily risk free rate returns. AR is the dummy variable for the event window (0,1). MKT represents market risk premium, while SMB (Small Minus Big) captures the return difference between small-cap and large-cap stocks. HML (High Minus Low) is a value factor and captures the difference between high book-to-market (value) stocks and low book-to-market (growth) stocks. MOM is a momentum factor that reflects the return difference between stocks with high past returns and those with low past returns. The estimation period begins 365 days before the event date (Aug 06, 2021). Column 1 provides excess returns for NYSE firms with all-male boards, while Column 2 provides excess returns for all firms listed on NYSE. Similarly, column 3 provides excess returns for NASDAQ firms with all-male board members (based on 2020 classification), and column 4 provides excess returns for all firms listed on NASDAQ. Standard errors are heteroskedasticity-robust.

Table 10: Market Reaction Analysis Using Portfolio Approach: Evidence from December 11, 2024

	(1) NYSE AMB	(2) NYSE	(3) NASDAQ AMB	(4) NASDAQ
Intercept	0.000	0.000^{*}	-0.000	0.000
	(0.001)	(0.000)	(0.001)	(0.000)
MKT	1.145^{***}	0.919^{***}	0.674^{***}	0.893^{***}
	(0.178)	(0.021)	(0.092)	(0.036)
SMB	0.582^{***}	0.531^{***}	0.558^{***}	0.918^{***}
	(0.180)	(0.022)	(0.093)	(0.035)
HML	0.343^{**}	0.426^{***}	-0.043	0.128^{***}
	(0.174)	(0.021)	(0.090)	(0.035)
MOM	-0.559^{***}	-0.171^{***}	-0.197^{**}	-0.162^{***}
	(0.182)	(0.022)	(0.094)	(0.036)
AR	-0.001	-0.002	-0.013^{*}	-0.006^{**}
	(0.013)	(0.002)	(0.007)	(0.003)
Observations	253	253	253	253
Adjusted R-squared	0.327	0.958	0.449	0.92

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table presents an analysis of stock returns based on daily portfolios, following the methodology of Eckbo, Nygaard, and Thorburn (2022). The dependent variable is the daily equally weighted average returns in excess of the daily risk free rate returns. AR is the dummy variable for the event window (0,1). MKT represents market risk premium, while SMB (Small Minus Big) captures the return difference between small-cap and large-cap stocks. HML (High Minus Low) is a value factor and captures the difference between high book-to-market (value) stocks and low book-to-market (growth) stocks. MOM is a momentum factor that reflects the return difference between stocks with high past returns and those with low past returns. The estimation period begins 365 days before the event date (December 11, 2024). Column 1 provides excess returns for NYSE firms with all-male boards, while Column 2 provides excess returns for all firms listed on the NYSE. Similarly, column 3 provides excess returns for NASDAQ firms with all-male board members (based on 2023 classification), and column 4 provides excess returns for all firms listed on NASDAQ. Standard errors are heteroskedasticity-robust.

Table 11: Market Reaction Analysis Using Portfolio Approach For NASDAQ Rule-Related Events

	(1) Dec 01, 2020	(2) Aug 10, 2021	(3) Oct 18, 2023	(4) Feb 19, 2024
Intercept	0.001***	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
MKT	0.918***	0.848***	0.775***	0.867***
	(0.014)	(0.032)	(0.030)	(0.033)
SMB	0.891***	0.944***	0.853***	0.817***
	(0.035)	(0.039)	(0.050)	(0.042)
HML	0.157***	−0.021	0.175***	0.165***
	(0.038)	(0.032)	(0.035)	(0.036)
MOM	−0.015	−0.054**	−0.145***	−0.183***
	(0.028)	(0.025)	(0.030)	(0.035)
AR	−0.001	−0.002	−0.005	0.004
	(0.003)	(0.003)	(0.003)	(0.004)
Observations	253	253	252	251
Adjusted R-squared	0.964	0.884	0.898	0.904

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table presents an analysis of stock returns based on daily portfolios constructed from all NASDAQ firms. The analysis follows the methodology of Eckbo, Nygaard, and Thorburn (2022) for NASDAQ rule-related events, excluding approval and vacatur dates. The variables and methodology are identical to those described in Table 9. Detailed descriptions of the underlying events are reported in Table A1. In the last column (Feb 19, 2024), the market was closed on February 19, 2024; therefore, the dummy variable AR is one on February 20, 2024, and zero otherwise.

Figure 1: Female Board Share

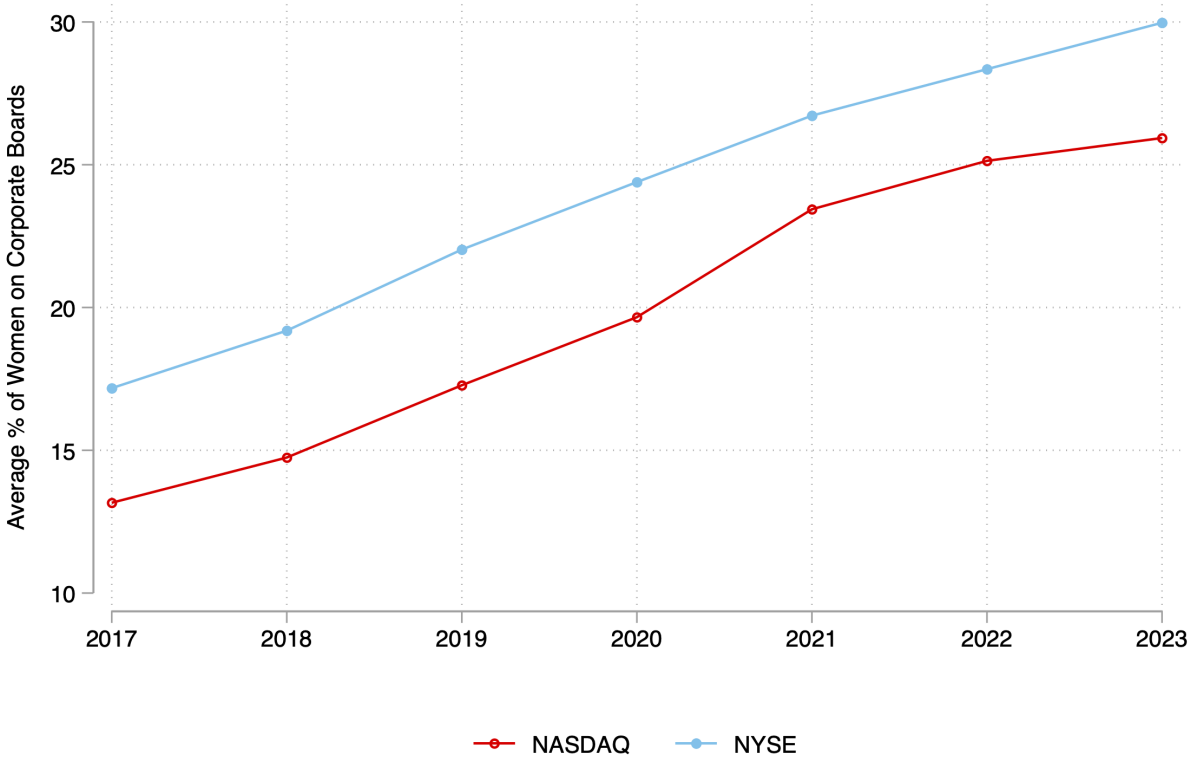
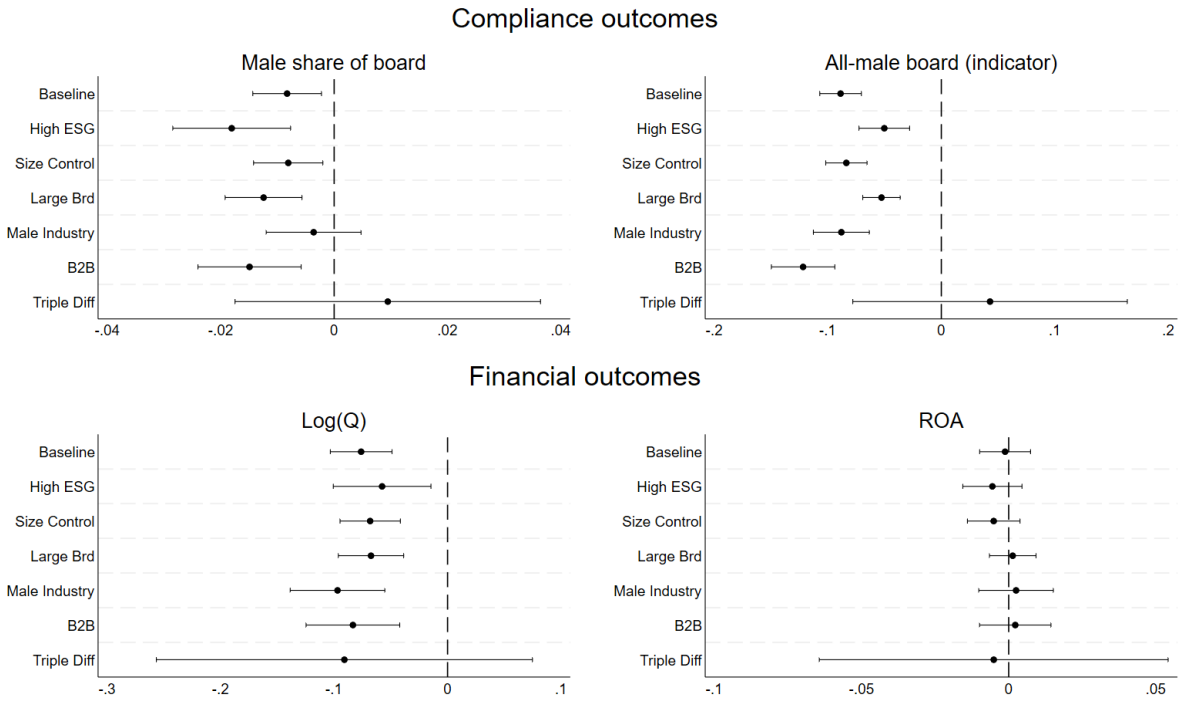


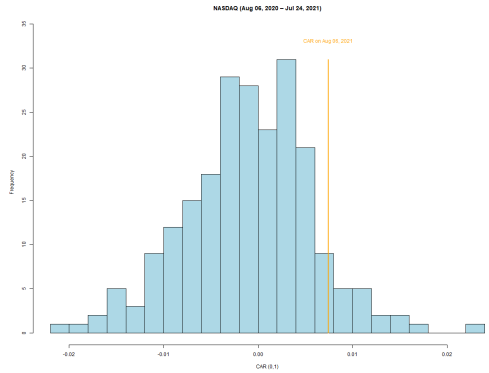
Figure 2: Effect of Mandatory Disclosure: Heterogeneity



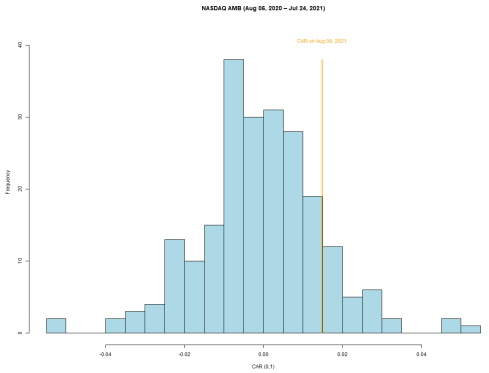
Notes: This figure reports reduced-form estimates of the NASDAQ board diversity disclosure rule on compliance and financial outcomes using an unbalanced panel of domestic listed firms, 2017–2023. Standard errors are clustered by firm. The baseline DiD compares NASDAQ (treated) to NYSE (control). Rows modify the baseline as follows: High ESG restricts to above-median MSCI industry-adjusted ESG firms; Size Control adds log(revenues); Large Board restricts to firms with >8 directors in 2020; Male Industry restricts to industries with below-average female board representation (2020 cross-section); B2B restricts to business-to-business firms; Triple Diff exploits pre/post variation across exchanges and whether boards were all-male in 2020. Board composition is from BoardEx organizational summary files (as of annual report date). GICS divisions (11) use 4-digit GICS from Compustat. Financials are from Compustat annual fundamentals and are log-transformed and/or winsorized at the 1st/99th percentiles using the domestic-listed annual distribution. The rule was SEC-approved August 6, 2021 and requires firms to have (or explain the absence of) at least two diverse directors by December 31, 2026 (one for boards with ≤ 5 directors).

Figure 3: Distribution of Placebo CARs Around the Approval and Vacatur Dates

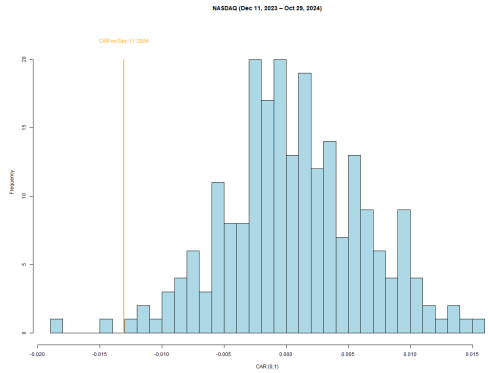
(a) Placebo CARs for NASDAQ firms around the approval date



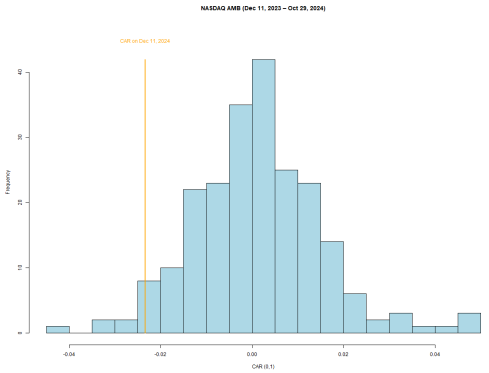
(b) Placebo CARs for NASDAQ AMB firms around the approval date



(c) Placebo CARs for NASDAQ firms around the vacatur date



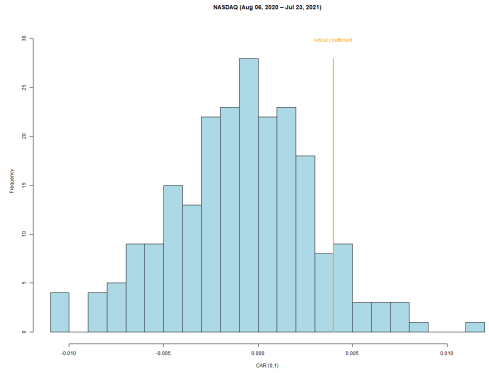
(d) Placebo CARs for NASDAQ AMB firms around the vacatur date



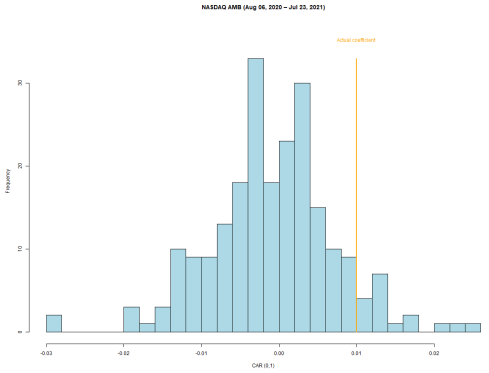
Notes: This figure presents the distribution of placebo cumulative abnormal returns (CARs) over the (0,+1) event window around the approval (Aug 06, 2021) and vacatur (Dec 11, 2024) dates. CARs are calculated using the standard event-study methodology with a 4-factor return model (Fama and French, 1993; Carhart, 1997), employing a 252 trading-day estimation window that ends 30 trading days prior to the event and requiring a minimum of 100 observations. Placebo event dates encompass all trading days between –252 and –30 trading days relative to each focal event date (i.e., August 06, 2021, and December 11, 2024). The yellow line represents the CAR on the approval and vacatur dates. AMB refers to an All-Male Board firm.

Figure 4: Distribution of Placebo Event Coefficients in Portfolio Return Regressions Around the Approval and Vacatur Dates

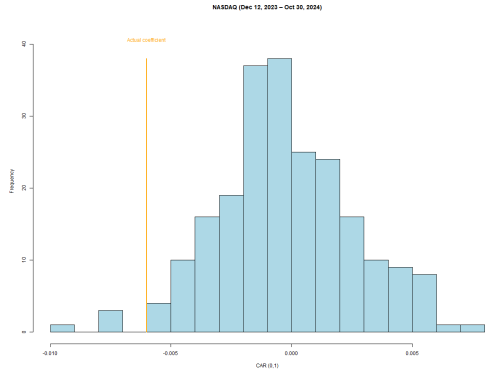
(a) Placebo indicator variable coefficients for portfolio returns of NASDAQ-listed firms around the approval date



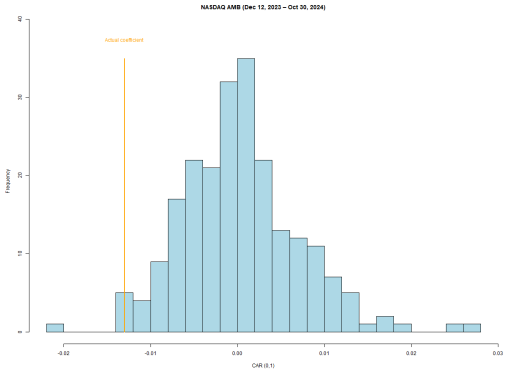
(b) Placebo indicator variable coefficients for portfolio returns of NASDAQ-listed AMB firms around the approval date



(c) Placebo indicator variable coefficients for portfolio returns of NASDAQ-listed firms around the vacatur date



(d) Placebo indicator variable coefficients for portfolio returns of NASDAQ-listed AMB firms around the vacatur date



Notes: This figure presents the distribution of placebo indicator variable coefficients for portfolio excess returns of NASDAQ-listed firms around the approval date (Aug 06, 2021) and vacatur date (Dec 11, 2024). The regression specification includes an indicator variable that takes the value one on the event day and the following trading day, and zero otherwise. The dependent variable is the daily excess return of an equally weighted portfolio composed of all NASDAQ firms and NASDAQ firms with all-male boards. The explanatory variables comprise the factors from the Fama-French three-factor model, the Carhart momentum factor, and the aforementioned event indicator. The estimation period spans 365 calendar days prior to the event through one day after the event. Placebo event dates encompass all trading days between –252 and –30 trading days relative to each focal event date (i.e., August 06, 2021, and December 11, 2024). The yellow line represents the indicator variable value on the approval and vacatur dates. AMB refers to an All-Male Board firm.

Figure 5: Textual Analysis of Explanations

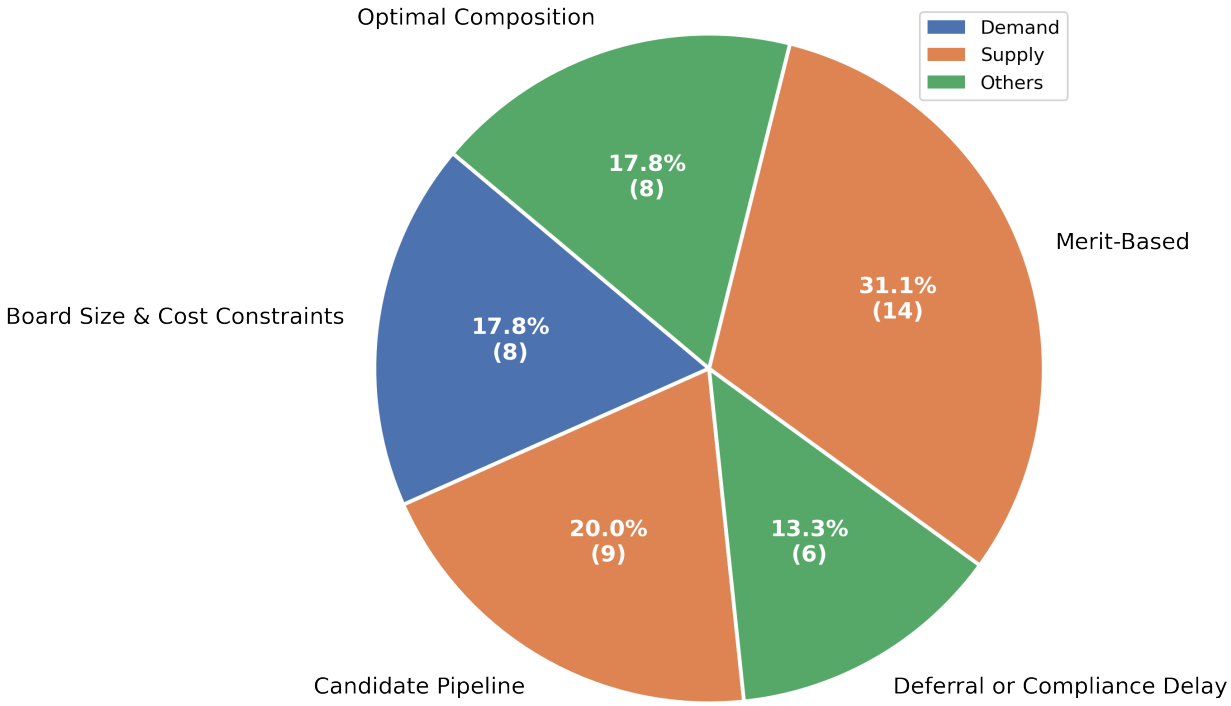
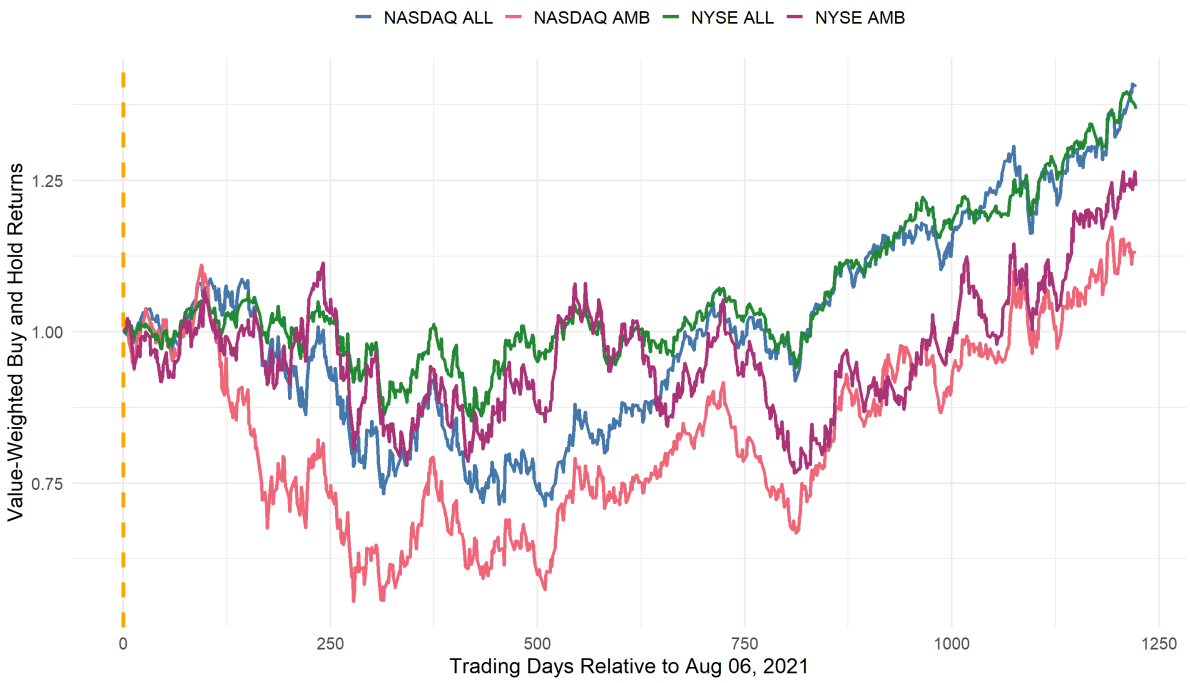


Figure 6: Value Weighted Buy and Hold Returns



Notes: The sample evaluates the performance of value-weighted portfolios of firms listed on NASDAQ and the NYSE. For firms delisted during the sample period, the returns on the day of delisting are computed as $(1+R_t)\times(1+DLRET_t)-1$, where R_t denotes the standard return on the day of delisting and $DLRET_t$ denotes the delisting return (Beaver, McNichols, and Price 2007). For these firms, post-delisting returns are set to zero. Firms with missing return data at any point between August 06, 2021, and December 10, 2024, are excluded from the analysis. Portfolios' weights are constructed using firms' market capitalization as of August 06, 2021. AMB refers to an All-Male Board firm.

Figure 7: SEC 14A Board Member Example

BOARD OF DIRECTORS INFORMATION

Biographical Information



Jeffrey P. Bezos

Founder, Chairman, and CEO of Amazon

Background

Mr. Bezos has been Chairman of the Board since founding the Company in 1994 and Chief Executive Officer since May 1996. Mr. Bezos served as President from founding until June 1999 and again from October 2000 to the present.

Qualifications and Skills

Mr. Bezos' individual qualifications and skills as a director include his customer-focused point of view, his willingness to encourage invention, his long-term perspective, and his on-going contributions as founder and CEO.

Age:	Director since:	Board committees:	Other current public company boards:
56	July 1994	None	None

Table A1: Timeline of NASDAQ's Board Diversity Rule

#	Event	Date	Description
1	NASDAQ proposed Board Diversity Rule	1-Dec-2020	NASDAQ submitted a rule proposal to the SEC requiring companies listed on its exchange to have one or two diverse directors—depending on the size of their board, or to publicly explain why they do not meet this criterion. The proposal also requires disclosure of board diversity statistics.
2	SEC approval (Final Rule Adopted)	6-Aug-2021	SEC approved the NASDAQ board diversity rule.
3	Petition filed for review in the Fifth Circuit challenging the SEC's approval of the rule	10-Aug-2021	The Alliance for Fair Board Recruitment challenged the SEC's approval of NASDAQ's diversity rule in the Fifth Circuit, arguing it exceeded the SEC's authority and violated constitutional and administrative law.
4	The Fifth Circuit upheld the rule	18-Oct-2023	A three-judge panel of the Fifth Circuit initially upheld the SEC's approval of the rule.
5	Deadline for at least one diverse director (or explain)	31-Dec-2023	NASDAQ-listed firms were required either to appoint at least one diverse director to their board or to publicly explain their inability to do so.
6	The Fifth Circuit granted a rehearing en banc	19-Feb-2024	The Fifth Circuit granted a rehearing en banc.
7	The Fifth Circuit (En Banc) Court struck down the rule	11-Dec-2024	The Fifth Circuit (en banc) issued a 9–8 decision vacating (invalidating) NASDAQ's board diversity rule.

Table A2: Attrition and IPO Rates

	Rate				Observations	
Year	NASDAQ	NYSE	Diff.	p-value	NASDAQ	NYSE
			Panel A: Attrition Rate			
2017	0.078	0.045	0.033***	0.000	1,996	1,209
2018	0.067	0.053	0.015*	0.095	2,019	1,196
2019	0.047	0.048	−0.001	0.941	2,044	1,178
2020	0.053	0.035	0.018**	0.022	2,157	1,162
2021	0.099	0.080	0.020**	0.042	2,638	1,333
2022	0.126	0.068	0.058***	0.000	2,554	1,290
			Panel B: IPO Rate			
2017	0.067	0.059	0.008	0.538	1,241	590
2018	0.078	0.037	0.041***	0.001	1,264	589
2019	0.074	0.031	0.043***	0.000	1,275	578
2020	0.098	0.050	0.048***	0.001	1,333	565
2021	0.169	0.118	0.051***	0.003	1,583	650
2022	0.037	0.006	0.031***	0.000	1,556	643
2023	0.025	0.013	0.012*	0.082	1,444	621

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Attrition rate is calculated between 2017 and 2022, and is done so separately for firms listed on the NASDAQ and NYSE. The attrition rate is equal to 1 if it is the last year that a firm is included in the dataset, and 0 otherwise. We omit 2023 from the attrition rate calculation because that is the last year of data availability for some variables. IPO rate is calculated between 2017 and 2023, and is done so separately for firms listed on the NASDAQ and NYSE. The IPO rate is equal to 1 if the IPO data for the firm is in that year, and 0 otherwise.

Table A3: BoardEx Matching Rates

Year	(1) BoardEx N	(2) CRSP	(3) Compustat	(4) All of (2-3)
2017	4,351	0.945	0.963	0.924
2018	4,337	0.942	0.963	0.923
2019	4,317	0.942	0.965	0.924
2020	4,507	0.935	0.955	0.911
2021	5,270	0.951	0.890	0.862
2022	5,009	0.966	0.925	0.902
2023	4,569	0.971	0.959	0.940
2024	3,717	0.978	0.976	0.961

Notes: Column (2) restricts to CRSP Company Policy and BoardEx matches, Column (3) restricts to Compustat Fundamentals and BoardEx, and Column (4) restricts to both.

Table A4: Variable Definitions and Data Sources

Variable	Description	Source
1(AMB)	Indicator for all male board	BoardEx
1(Expand)	Indicator for increase in board size relative to previous year	BoardEx
Board Size	Board size	BoardEx
Female Directors	Number of female directors	BoardEx
MBA	Percent of board with an MBA degree	BoardEx
Age	Time since IPO date	Compustat
Number of Employees	Number of employees at firm in thousands	Compustat
Tobin's Q	(CSHO*PRCC + DLTT + DLC)/AT	Compustat
Total Assets	Compustat item AT	Compustat
Sales	Compustat item SALE	Compustat
ROA	Compustat item NI/AT	Compustat
Leverage	Compustat item (DLC+DLTT)/AT	Compustat
Cash to Assets	Compustat item CHE/AT	Compustat
Market Value	CRSP item shrout*prc	CRSP
1(Merger)	Indicator for merger	CRSP
1(NASDAQ)	Indicator for NASDAQ-listed firms	CRSP
1(Dividend)	Indicator for dividend	CRSP
1(Dec in Shares Outstanding $\geq 5\%$)	Indicator for decrease in shares outstanding	CRSP
1(Inc in Shares Outstanding $\geq 5\%$)	Indicator for increase in shares outstanding	CRSP
Abnormal Returns	Abnormal returns are computed as the difference between firms' returns obtained from CRSP and their expected returns. Expected returns are estimated using the Carhart (1997) model. The Fama–French and momentum factors are obtained from WRDS	CRSP, WRDS
Sustainalytics ESG Score	Sustainalytics ESG Risk Score	Sustainalytics
MSCI ESG Score	MSCI ESG Weighted Average Score	MSCI

Table A5: AMB Summary Statistics in 2020

	Mean				Observations	
	NASDAQ	NYSE	Diff.	p-value	NASDAQ	NYSE
Board Characteristics						
Board Size	5.94	6.16	−0.22	0.380	349	44
% Directors with MBA	0.30	0.29	0.02	0.706	273	36
% Directors with Doctoral Degree	0.11	0.05	0.06*	0.070	273	36
% Directors with Ivy League Degree	0.21	0.25	−0.04	0.369	273	36
Average Director Tenure	0.05	0.05	−0.00	0.972	273	36
% Directors with Board Exp	0.82	0.84	−0.02	0.623	273	36
% Directors with C-Suite Exp	0.67	0.61	0.06	0.213	273	36
% Directors with Sector Exp	0.64	0.52	0.12*	0.055	273	36
1(AMB)	1.00	1.00	0.00	.	349	44
1(Expand)	0.10	0.16	−0.06	0.264	349	44
Female Ratio	0.00	0.00	0.00	.	349	44
Age	12.60	16.84	−4.23	0.139	170	17
Number of Employees (thousands)	0.72	2.37	−1.65***	0.000	304	36
Business-to-Business	0.72	0.72	−0.00	0.961	253	25
Business-to-Consumer	0.28	0.28	0.00	0.961	253	25
Firm Characteristics						
Return on Assets	−0.10	−0.05	−0.06	0.390	317	38
Log(Tobin's Q)	0.53	0.13	0.40**	0.049	301	35
Total Assets ($ mill)	439.39	1,447.15	−1,007.76***	0.000	318	38
Sales ($ mill)	45.04	232.09	−187.05***	0.000	317	38
Leverage	0.21	0.30	−0.09	0.191	309	38
Cash to Assets	0.41	0.21	0.20***	0.000	318	38
Bid-Ask Spread	0.02	0.01	0.01**	0.047	349	44
Company Policy						
1(Merger)	0.00	0.02	−0.02***	0.005	349	44
1(Dividend)	0.15	0.36	−0.22***	0.000	349	44
1(Dec in Shares Outstanding $\geq$ 5%)	0.13	0.16	−0.03	0.619	349	44
1(Incr in Shares Outstanding $\geq$ 5%)	0.08	0.05	0.04	0.384	349	44
ESG						
Sustainalytics ESG Score	31.50	31.42	0.08	0.971	79	22
MSCI ESG Score	4.20	3.88	0.32	0.318	38	11
1(ESG High)	0.26	0.27	−0.01	0.951	38	11
Industry						
Energy	0.03	0.16	−0.13***	0.000	349	44
Materials	0.01	0.05	−0.03	0.142	349	44
Industrials	0.10	0.18	−0.08	0.103	349	44
Consumer Discretionary	0.09	0.14	−0.04	0.383	349	44
Consumer Staples	0.03	0.02	0.01	0.823	349	44
Health Care	0.34	0.02	0.32***	0.000	349	44
Financials	0.13	0.05	0.09*	0.092	349	44
Information Technology	0.16	0.11	0.05	0.420	349	44
Communication Services	0.05	0.07	−0.02	0.516	349	44
Utilities	0.00	0.02	−0.02*	0.081	349	44
Real Estate	0.01	0.14	−0.12***	0.000	349	44

Continued on next page

(Table A5 continued)

	Mean				Observations	
	NASDAQ	NYSE	Diff.	*p*-value	NASDAQ	NYSE

Notes: The sample restricts to all-male boards. The table comes from a cross section of the data, representing 2020 values. Raw means and p-values from a two-sided t-test are reported. Boardroom characteristics are derived from BoardEx and represent mean values in 2020. Firm characteristics are derived from Compustat and represent mean values in 2020. Return on Assets is net income before extraordinary items and discontinued operations divided by book assets. Tobin's Q is the ratio of the firm's market value to its book value of assets. Total Assets refers to Compustat item AT. Sales refers to Compustat item SALE. Leverage refers to book value of long- and short-term debt divided by total assets. Cash to Assets refers to cash and short-term investments divided by total assets. All company policies indicate if the event occurred for some security during the calendar year, and are derived from CRSP's Events files. Codes for industry classification are also derived from Compustat.

Table A6: Effect of Mandatory Disclosure on Board Composition: Already-Diverse Firms

Dependent variable:	Male Share of Board (1)	1(All-Male Board) (2)	Board Size (3)	1(Expand Board) (4)	1(Male Dropped) (5)
	Panel A: Intent to Treat Effect Estimates				
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.009^{***}	-0.061^{***}	-0.061	0.012	-0.009
	(0.003)	(0.008)	(0.047)	(0.013)	(0.008)
	Panel B: Event Studies Estimates				
NASDAQ $\times$ Year = 2017	0.011^{***}	0.129^{***}	0.047	0.000	-0.027^{*}
	(0.004)	(0.016)	(0.061)	(0.026)	(0.015)
NASDAQ $\times$ Year = 2018	0.013^{***}	0.106^{***}	0.075	0.028	-0.030^{**}
	(0.004)	(0.013)	(0.056)	(0.027)	(0.015)
NASDAQ $\times$ Year = 2019	0.008^{***}	0.039^{***}	0.069	0.040	-0.019
	(0.003)	(0.008)	(0.046)	(0.029)	(0.016)
NASDAQ $\times$ Year = 2021	-0.002	0.004	-0.019	0.023	-0.028^{*}
	(0.003)	(0.004)	(0.045)	(0.028)	(0.016)
NASDAQ $\times$ Year = 2022	-0.004	0.002	0.014	0.044^{*}	-0.019
	(0.003)	(0.005)	(0.055)	(0.026)	(0.015)
NASDAQ $\times$ Year = 2023	0.001	0.006	-0.040	0.019	-0.036^{**}
	(0.004)	(0.005)	(0.059)	(0.025)	(0.015)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year-SIC FE	Yes	Yes	Yes	Yes	Yes
Observations	23,320	23,320	23,320	21,695	21,695
Dependent variable mean	0.766	0.072	8.854	0.272	0.069
Number of firms	4,181	4,181	4,181	3,948	3,948

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the firm level are reported in parentheses. The sample restricts to an unbalanced panel of firms that were domestic, listed, and did not have an all-male-board in 2020 (already-diverse), covering time years 2017-2023. Panel A reports difference-in-differences specifications with firm and Year-SIC fixed effects. Panel B reports event studies estimates relative to the 2020 baseline. Standard errors are clustered at the firm level. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. The outcome variables are derived from BoardEx's organizational summary files, which provide the director roster as of the company's annual report date. The 'Expand Board' indicator equals one if board size increases relative to the prior year. 'Male Dropped' equals 1 if board size is unchanged relative to the prior year while the proportion of male directors fall. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table A7: Effect of Mandatory Disclosure on Board Composition - Synthetic DiD

Dependent variable:	Male Share of Board (1)	1(All-Male Board) (2)	Board Size (3)	1(Expand Board) (4)	1(Male Dropped) (5)
Panel A: Synthetic Difference-in-Differences Intent to Treat Effect Estimates					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.008^{***}	-0.045^{***}	0.023	0.016	0.004
	(0.003)	(0.007)	(0.041)	(0.012)	(0.008)
Panel B: Difference-in-Differences Intent to Treat Effect Estimates (on Balanced Panels)					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.009^{***}	-0.089^{***}	-0.064	0.031^{**}	0.007
	(0.003)	(0.010)	(0.049)	(0.013)	(0.009)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	18,410	18,410	18,410	17,381	17,381
Dependent variable mean	0.779	0.118	8.949	0.262	0.065
Number of firms	2,630	2,630	2,630	2,428	2,428

Notes: Jackknife standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The samples restrict to balanced panels (required for synthetic difference-in-difference) of firms that were domestic and listed covering 2017–2023. Panel A shows results from a synthetic difference-in-difference specification following Arkhangelsky et al., (2021). Panel B shows comparison of conventional difference-in-difference results on the same balanced panels. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. The outcome variables are derived from BoardEx's organizational summary files, which provide the director roster as of the company's annual report date. The 'Expand Board' indicator equals one if board size increases relative to the prior year. 'Male Dropped' equals 1 if board size is unchanged relative to the prior year while the proportion of male directors fall. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table A8: Effect of Mandatory Disclosure on Financial Variables

Dependent variable:	RoA	RoE	Tobin's Q	Log(Market to Book)	Cash Flow to Asset	Relative Spread (%)
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Intent to Treat Effect Estimates					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.001	-0.018	-0.217^{***}	-0.103^{***}	-0.002	-0.002^{***}
	(0.004)	(0.019)	(0.0143)	(0.024)	(0.004)	(0.000)
	Panel B: DiD Event Studies Estimates					
NASDAQ $\times$ Year = 2017	-0.022^{***}	-0.016	-0.139^{*}	-0.023	-0.019^{**}	-0.001^{***}
	(0.007)	(0.028)	(0.071)	(0.030)	(0.008)	(0.000)
NASDAQ $\times$ Year = 2018	-0.029^{***}	-0.054^{**}	-0.246^{***}	-0.050^{*}	-0.027^{***}	-0.001^{*}
	(0.008)	(0.027)	(0.070)	(0.028)	(0.008)	(0.000)
NASDAQ $\times$ Year = 2019	-0.025^{***}	-0.065^{**}	-0.203^{***}	-0.064^{***}	-0.022^{***}	-0.000
	(0.007)	(0.026)	(0.062)	(0.024)	(0.007)	(0.000)
NASDAQ $\times$ Year = 2021	0.012^{*}	0.037	-0.253^{***}	-0.083^{***}	0.012^{*}	-0.003^{***}
	(0.007)	(0.025)	(0.068)	(0.026)	(0.007)	(0.000)
NASDAQ $\times$ Year = 2022	-0.025^{***}	-0.083^{**}	-0.427^{***}	-0.164^{***}	-0.024^{***}	-0.002^{***}
	(0.008)	(0.035)	(0.079)	(0.032)	(0.009)	(0.000)
NASDAQ $\times$ Year = 2023	-0.048^{***}	-0.117^{***}	-0.413^{***}	-0.168^{***}	-0.046^{***}	0.006^{***}
	(0.009)	(0.035)	(0.080)	(0.034)	(0.009)	(0.000)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-SIC FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,453	23,975	22,725	22,726	24,122	21,532
Dependent variable mean	0.783	0.133	2.178	0.266	0.066	0.006
Number of firms	4,546	4,382	4,307	4,307	4,347	3,907

Notes: Clustered (firm) standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The sample restricts to an unbalanced panel of firms that were domestic and listed covering 2017–2023. Panel A shows estimates of difference-in-difference average treatment effect. Panel B reports event studies estimates relative to the 2020 baseline. Standard errors are clustered at the firm level. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. Financial variables are derived from Compustat's quarterly fundamental files and are either log transformed or winsorized at the 1st and 99th percentiles. The percentiles are calculated relative to all domestic and listed companies observed in the annual distribution. The index of financial outcomes averages the z-score across several financial outcomes, following Kling, Liebman, and Katz (2007). For each financial outcome, the z-score subtracts the mean of the control group, then divides by the standard deviation of the control group. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table A9: Effect of Mandatory Disclosure on Financial Variables

Dependent variable:	RoA	RoE	Log(Q)	Log(Market to Book)	Cash Flow to Asset	Relative Spread (%)
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Synthetic Difference-in-Difference					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	-0.004	-0.022	-0.107^{***}	-0.134^{***}	-0.007	-0.0009^{***}
	(0.012)	(0.021)	(0.015)	(0.023)	(0.011)	(0.0002)
	Panel B: Difference-in-Difference (Balanced Panels)					
NASDAQ $\times \mathbb{I}$(Year $>$ 2020)	0.002	-0.007	-0.043^{***}	-0.058^{**}	0.001	-0.0009^{***}
	(0.005)	(0.018)	(0.015)	(0.026)	(0.005)	(0.0002)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,850	15,799	13,755	13,755	16,660	14,960
Dependent variable mean	-0.037	-0.006	0.507	0.843	-0.01	0.005
Number of firms	2,550	2,257	1,965	1,965	2,380	2,105

Notes: Jackknife standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The samples restrict to balanced panels (required for synthetic difference-in-difference) of firms that were domestic and listed covering 2017–2023. Panel A shows results from a synthetic difference-in-difference specification following Arkhangelsky et al., (2021). Panel B shows comparison of conventional difference-in-difference results on the same balanced panels. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. Financial variables are derived from Compustat's annual fundamental files and are either log transformed or winsorized at the 1st and 99th percentiles. The percentiles are calculated relative to all domestic and listed companies observed in the annual distribution. The index of financial outcomes averages the z-score across several financial outcomes, following Kling, Liebman, and Katz (2007). For each financial outcome, the z-score subtracts the mean of the control group, then divides by the standard deviation of the control group. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table A10: Effect of Mandatory Disclosure on Financial Variables

Dependent variable:	RoA	RoE	Log(Q)	Log(Market to Book)	Cash Flow to Asset	Relative Spread (%)
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Synthetic DiD Event Studies Estimates					
NASDAQ × Year = 2017	0.001	0.016	−0.003	−0.003	0.001	0.0002
	(0.003)	(0.010)	(0.005)	(0.006)	(0.003)	(0.0002)
NASDAQ × Year = 2018	−0.006***	−0.010	−0.019***	−0.006	−0.007***	−0.0002
	(0.002)	(0.006)	(0.006)	(0.007)	(0.002)	(0.0001)
NASDAQ × Year = 2019	−0.002	−0.012	−0.020***	−0.005	−0.001	0.0001
	(0.003)	(0.010)	(0.006)	(0.006)	(0.003)	(0.0001)
NASDAQ × Year = 2021	0.026***	0.047***	−0.062***	−0.087***	0.024**	−0.002***
	(0.009)	(0.017)	(0.013)	(0.021)	(0.010)	(0.0002)
NASDAQ × Year = 2022	−0.007	−0.037	−0.134***	−0.161***	−0.014	−0.0007***
	(0.013)	(0.026)	(0.017)	(0.029)	(0.013)	(0.0002)
NASDAQ × Year = 2023	−0.031**	−0.075**	−0.126***	−0.154***	−0.032***	0.0004
	(0.012)	(0.032)	(0.019)	(0.032)	(0.012)	(0.0002)
	Panel B: Synthetic DiD Estimates (ATT)					
NASDAQ × $\mathbb{I}$(Year > 2020)	−0.004	−0.022	−0.107***	−0.134***	−0.007	−0.0009***
	(0.010)	(0.019)	(0.014)	(0.024)	(0.010)	(0.0002)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-SIC FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,850	15,799	13,755	13,755	16,660	14,960
Dependent variable mean	−0.037	−0.006	0.507	0.843	−0.01	0.005
Number of firms	2,550	2,257	1,965	1,965	2,380	2,105

Notes: Clustered (firm) standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The samples restrict to balanced panels (required for synthetic difference-in-difference) of firms that were domestic and listed covering 2017–2023. Panel A reports event studies estimates relative to the synthetic control trend constructed to match pre-treatment outcomes. Panel B shows estimates of synthetic difference-in-difference average treatment effect for comparison. Standard errors are bootstrapped based on 999 simulations and clustered at the firm level. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. Financial variables are derived from Compustat's quarterly fundamental files and are either log transformed or winsorized at the 1st and 99th percentiles. The percentiles are calculated relative to all domestic and listed companies observed in the annual distribution. The index of financial outcomes averages the z-score across several financial outcomes, following Kling, Liebman, and Katz (2007). For each financial outcome, the z-score subtracts the mean of the control group, then divides by the standard deviation of the control group. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table A11: Effect of Board Member Composition on Financial Outcomes

	(1) NASDAQ	(2) NYSE
Panel A: Instrumental Variable regressions: dependent var = financial variables		
industry-adjusted Q		
Percent women directors	−1.313	−2.578
	(2.410)	(2.389)
industry-adjusted RoA		
Percent women directors	−0.177	0.154
	(0.335)	(0.206)
industry-adjusted RoE		
Percent women directors	−1.133***	2.113*
	(0.393)	(1.140)
Log of industry-adjusted Market to Book		
Percent women directors	0.343	0.736
	(1.354)	(2.259)
Industry-adjusted Cash flow to Asset		
Percent women directors	0.002	0.124
	(0.326)	(0.206)
Year FE	Yes	Yes
Firm FE	Yes	Yes
Panel B: Reduced-form regressions: dependent var = industry-adjusted Q		
2021 dummy × percent of women in 2020	0.616	0.325
	(0.773)	(0.569)
2022 dummy × percent of women in 2020	0.335	0.308
	(0.845)	(0.693)
2023 dummy × percent of women in 2020	0.583	0.970
	(0.845)	(0.761)
Year FE	Yes	Yes
Firm FE	Yes	Yes
F-statistic	0.86	1.40
Observations	6,740	4,092
Panel C: First-stage regressions: dependent var = percent women directors$_t$		
2021 dummy	0.069***	0.063***
	(0.004)	(0.005)
2022 dummy	0.110***	0.103***
	(0.005)	(0.007)
2023 dummy	0.135***	0.136***
	(0.005)	(0.007)
2021 dummy × percent of women in 2020	−0.163***	−0.157***
	(0.015)	(0.019)
2022 dummy × percent of women in 2020	−0.288***	−0.252***
	(0.020)	(0.024)
2023 dummy × percent of women in 2020	−0.355***	−0.327***
	(0.023)	(0.027)
Firm FE	Yes	Yes
F-statistic	169.23	102.55
Observations	6,918	4,270

Notes: Clustered (firm) standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Data are yearly observations from 2020 to 2023. Percent women directors$_t$ is a predicted variable estimated in the first-stage regressions. Percent of women in 2020 records the percentage of shareholder-elected directors that were women as reported in the firm's 2020 annual report. Year-2020 variables are omitted. Standard errors are clustered by firm. Financial variables in Panel A are industry-adjusted by subtracting the median for the same four-digit GICS industry among firms pooled across NASDAQ & NYSE. If fewer than five firms exist at the four-digit level, the three-digit industry-group median is used; if still fewer than five, the two-digit sector median is used.

Table A12: Abnormal Returns on December 11, 2024 for NASDAQ's Explaining and Non-Explaining Firms

				Tests of mean = 0			
	Day relative to event	No. of firms	Mean (%)	Std. cross-sectional t-test	Patell's Z test	Gen. sign test	Wilcoxon signed-rank test
Non-Explaining	0	2,075	−0.743	***	***	***	***
	1		−0.577	***	***	**	***
Explaining	0	41	0.079				*
	1		−0.719				

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The mean on day 0 represents the average abnormal returns on the event date, while the mean on day 1 reflects the average abnormal returns on the day following the event date. For the abnormal returns, we employed an estimation window of 252 days, included a gap of 30 days between the end of the estimation window and the event date (Dec 11, 2024), and only considered firms with at least 100 stock return observations in the estimation window. Explaining firms are those that offered an explanation for their lack of diversity.

Table A13: Effect of Mandatory Disclosure on Board Composition - Synthetic DiD Event Studies

Dependent variable:	Male Share of Board (1)	1(All-Male Board) (2)	Board Size (3)	1(Expand Board) (4)	1(Male Dropped) (5)
Panel A: Synthetic Difference-in-Differences Event Studies Estimates					
NASDAQ × Year = 2017	−0.000	0.011***	0.001	−0.004	0.000
	(0.000)	(0.003)	(0.006)	(0.004)	(0.002)
NASDAQ × Year = 2018	0.002***	0.015***	0.003	0.002	−0.006**
	(0.001)	(0.003)	(0.008)	(0.004)	(0.003)
NASDAQ × Year = 2019	0.002**	0.006	0.008	0.006	−0.002
	(0.001)	(0.004)	(0.008)	(0.005)	(0.003)
NASDAQ × Year = 2021	−0.009***	−0.032***	−0.025	0.003	0.005
	(0.003)	(0.006)	(0.041)	(0.021)	(0.012)
NASDAQ × Year = 2022	−0.009***	−0.046***	0.050	0.049**	0.008
	(0.003)	(0.008)	(0.048)	(0.020)	(0.011)
NASDAQ × Year = 2023	−0.007*	−0.056***	0.045	−0.005	−0.000
	(0.004)	(0.009)	(0.053)	(0.018)	(0.010)
Panel B: Synthetic Difference-in-Differences Intent to Treat Effect Estimates					
NASDAQ × $\mathbb{I}$(Year > 2020)	−0.008***	−0.045***	0.023	0.016	0.004
	(0.003)	(0.007)	(0.040)	(0.013)	(0.008)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year-SIC FE	Yes	Yes	Yes	Yes	Yes
Observations	18,410	18,410	18,410	17,381	17,381
Dependent variable mean	0.779	0.118	8.949	0.262	0.065
Number of firms	2,630	2,630	2,630	2,428	2,428

Notes: Clustered (firm) standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The samples restrict to balanced panels (required for synthetic difference-in-difference) of firms that were domestic and listed covering 2017–2023. Panel A reports event studies estimates relative to the synthetic control trend constructed to match pre-treatment outcomes. Panel B shows estimates of synthetic difference-in-difference average treatment effect for comparison. Standard errors are bootstrapped based on 999 simulations and clustered at the firm level. Treated firms are listed in NASDAQ. Control firms are listed in NYSE. The outcome variables are derived from BoardEx's organizational summary files, which provide the director roster as of the company's annual report date. The 'Expand Board' indicator equals one if board size increases relative to the prior year. 'Male Dropped' equals 1 if board size is unchanged relative to the prior year while the proportion of male directors fall. Industries are categorized into 11 divisions using the 4-digit GICS code. GICS codes are obtained from Compustat. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.

Table A14: Effects of Mandatory Disclosure on Board Composition of Firms with All Male Boards

	Male Share of Board	1(All-Male Board)	Board Size	1(Expand Board)	1(Male Dropped)
NASDAQ × Year = 2017	0.006	−0.041	−0.364	−0.061	0.004
	(0.012)	(0.057)	(0.363)	(0.115)	(0.010)
NASDAQ × Year = 2018	0.004	−0.054	−0.483	0.012	0.009
	(0.014)	(0.062)	(0.350)	(0.095)	(0.009)
NASDAQ × Year = 2019	0.011	0.005	−0.255	0.065	0.008
	(0.013)	(0.055)	(0.275)	(0.099)	(0.009)
NASDAQ × Year = 2021	0.028	0.165*	−0.653**	−0.172	0.013
	(0.019)	(0.094)	(0.298)	(0.109)	(0.048)
NASDAQ × Year = 2022	0.016	0.010	−0.443	0.178**	0.039
	(0.022)	(0.093)	(0.281)	(0.081)	(0.034)
NASDAQ × Year = 2023	0.033	−0.009	−0.272	−0.032	0.005
	(0.024)	(0.090)	(0.321)	(0.110)	(0.039)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year-SIC FE	Yes	Yes	Yes	Yes	Yes
Observations	1,856	1,856	1,856	1,720	1,720
Dependent variable mean	0.949	0.730	6.379	0.214	0.035
Number of firms	313	313	313	310	310

Notes: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The sample restricts to an unbalanced panel of firms that were domestic, listed, and had all-male boards in 2020. The time period covered is 2017 - 2023, with reported effects relative to the 2020 baseline. Standard errors are clustered at the firm level. Treated firms are listed in NASDAQ and have all-male boards as of 2020. Control firms are listed in NYSE and have all-male boards as of 2020. The 'Expand Board' indicator equals one if board size increases relative to the prior year. 'Male Dropped' equals 1 if some male director present in the previous year is not present in the current year. Outcome variables related to board composition are derived from BoardEx's organizational summary files, which provide the director roster as of the company's annual report date. Industries are categorized into 11 divisions using the 4 digit GICS code. GICS codes are obtained from Compustat. Sample sizes vary due to missing values of the outcome variable. The NASDAQ board diversity rule was approved on August 6, 2021, mandating companies to have at least two diverse directors on board (at least one for companies with five or fewer directors by December 31, 2026) or explain why they do not comply.